

- (d) The range of A is F^n .
- (e) The equation $A\mathbf{x} = \mathbf{b}$ is consistent for each $\mathbf{b} \in F^n$.
- (f) If the equation $A\mathbf{x} = \mathbf{b}$ is consistent, then the solution is unique.
- (g) The equation $A\mathbf{x} = \mathbf{b}$ has a unique solution for each $\mathbf{b} \in F^n$.
- (h) The rows of A are linearly independent.
- (i) The dimension of $\text{RS}(A)$ is n .
- (j) The row space of A is F^n .
- (k) The dimension of $\ker(A)$ is 0.
- (l) The only solution to $A\mathbf{x} = \mathbf{0}$ is $\mathbf{x} = \mathbf{0}$.
- (m) The rank of A is n .
- (n) The determinant of A is nonzero. (See Section 4.1 for the definition of the determinant.)

2.6 Coordinates and Change of Basis

Coordinates are used to transform a problem in a more abstract vector space (e.g., the vector space of polynomials of degree less than or equal to 3) to a problem in F^n .

Definitions:

Suppose that $\mathcal{B} = (\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n)$ is an ordered basis for a vector space V over a field F and $\mathbf{x} \in V$. The **coordinates of $\mathbf{x}$ relative to the ordered basis $\mathcal{B}$** (or the **$\mathcal{B}$ -coordinates of $\mathbf{x}$**) are the scalar coefficients $c_1, c_2, \dots, c_n \in F$ such that $\mathbf{x} = c_1\mathbf{x}_1 + c_2\mathbf{x}_2 + \dots + c_n\mathbf{x}_n$. Whenever coordinates are involved, the vector space is assumed to be nonzero and finite dimensional.

If $c_1, c_2, \dots, c_n$ are the $\mathcal{B}$ -coordinates of $\mathbf{x}$, then the vector in F^n ,

$$[\mathbf{x}]_{\mathcal{B}} = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix},$$

is the **coordinate vector of $\mathbf{x}$ relative to $\mathcal{B}$** or the **$\mathcal{B}$ -coordinate vector of $\mathbf{x}$** .

The mapping $\mathbf{x} \rightarrow [\mathbf{x}]_{\mathcal{B}}$ is the **coordinate mapping determined by $\mathcal{B}$** .

If $\mathcal{B}$ and $\mathcal{B}'$ are ordered bases for the vector space F^n , then the **change-of-basis matrix** from $\mathcal{B}$ to $\mathcal{B}'$ is the matrix whose columns are the $\mathcal{B}'$ -coordinate vectors of the vectors in $\mathcal{B}$ and is denoted by ${}_{\mathcal{B}'}[I]_{\mathcal{B}}$. Such a matrix is also called a **transition matrix**.

Facts: The following facts can be found in [Lay03, Sections 4.4 and 4.7] or [HJ85, Section 0.10]:

- For any vector $\mathbf{x} \in F^n$ with the standard ordered basis $\mathcal{E}_n = (\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n)$, we have $\mathbf{x} = [\mathbf{x}]_{\mathcal{E}_n}$.
- For any ordered basis $\mathcal{B} = (\mathbf{b}_1, \dots, \mathbf{b}_n)$ of a vector space V , we have $[\mathbf{b}_i]_{\mathcal{B}} = \mathbf{e}_i$.
- If $\dim(V) = n$, then the coordinate mapping is a one-to-one linear transformation from V onto F^n . (See Chapter 3 for the definition of linear transformation.)
- If $\mathcal{B}$ is an ordered basis for a vector space V and $\mathbf{v}_1, \mathbf{v}_2 \in V$, then $\mathbf{v}_1 = \mathbf{v}_2$ if and only if $[\mathbf{v}_1]_{\mathcal{B}} = [\mathbf{v}_2]_{\mathcal{B}}$.
- Let V be a vector space over a field F , and suppose $\mathcal{B}$ is an ordered basis for V . Then for any $\mathbf{x}, \mathbf{v}_1, \dots, \mathbf{v}_k \in V$ and $c_1, \dots, c_k \in F$, $\mathbf{x} = c_1\mathbf{v}_1 + \dots + c_k\mathbf{v}_k$ if and only if $[\mathbf{x}]_{\mathcal{B}} = c_1[\mathbf{v}_1]_{\mathcal{B}} + \dots + c_k[\mathbf{v}_k]_{\mathcal{B}}$. So, for any $\mathbf{x}, \mathbf{v}_1, \dots, \mathbf{v}_k \in V$, $\mathbf{x} \in \text{Span}(\mathbf{v}_1, \dots, \mathbf{v}_k)$ if and only if $[\mathbf{x}]_{\mathcal{B}} \in \text{Span}([\mathbf{v}_1]_{\mathcal{B}}, \dots, [\mathbf{v}_k]_{\mathcal{B}})$.
- Suppose $\mathcal{B}$ is an ordered basis for an n -dimensional vector space V over a field F and $\mathbf{v}_1, \dots, \mathbf{v}_k \in V$. The set $S = \{\mathbf{v}_1, \dots, \mathbf{v}_k\}$ is linearly independent in V if and only if the set $S' = \{[\mathbf{v}_1]_{\mathcal{B}}, \dots, [\mathbf{v}_k]_{\mathcal{B}}\}$ is linearly independent in F^n .

7. Let V be a vector space over a field F with $\dim(V) = n$, and suppose $\mathcal{B}$ is an ordered basis for V . Then $\text{Span}(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k) = V$ for some $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k \in V$ if and only if $\text{Span}([\mathbf{v}_1]_{\mathcal{B}}, [\mathbf{v}_2]_{\mathcal{B}}, \dots, [\mathbf{v}_k]_{\mathcal{B}}) = F^n$.
8. Suppose $\mathcal{B}$ is an ordered basis for a vector space V over a field F with $\dim(V) = n$, and let $S = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ be a subset of V . Then S is a basis for V if and only if $\{[\mathbf{v}_1]_{\mathcal{B}}, \dots, [\mathbf{v}_n]_{\mathcal{B}}\}$ is a basis for F^n if and only if the matrix $[[\mathbf{v}_1]_{\mathcal{B}}, \dots, [\mathbf{v}_n]_{\mathcal{B}}]$ is invertible.
9. If $\mathcal{B}$ and $\mathcal{B}'$ are ordered bases for a vector space V , then $[\mathbf{x}]_{\mathcal{B}'} = {}_{\mathcal{B}'}[I]_{\mathcal{B}} [\mathbf{x}]_{\mathcal{B}}$ for any $\mathbf{x} \in V$. Furthermore, ${}_{\mathcal{B}'}[I]_{\mathcal{B}}$ is the only matrix such that for any $\mathbf{x} \in V$, $[\mathbf{x}]_{\mathcal{B}'} = {}_{\mathcal{B}'}[I]_{\mathcal{B}} [\mathbf{x}]_{\mathcal{B}}$.
10. Any change-of-basis matrix is invertible.
11. If B is invertible, then B is a change-of-basis matrix. Specifically, if $B = [\mathbf{b}_1 \cdots \mathbf{b}_n] \in F^{n \times n}$, then $B = {}_{\mathcal{E}_n}[I]_{\mathcal{B}}$, where $\mathcal{B} = (\mathbf{b}_1, \dots, \mathbf{b}_n)$ is an ordered basis for F^n .
12. If $\mathcal{B} = (\mathbf{b}_1, \dots, \mathbf{b}_n)$ is an ordered basis for F^n , then ${}_{\mathcal{E}_n}[I]_{\mathcal{B}} = [\mathbf{b}_1 \cdots \mathbf{b}_n]$.
13. If $\mathcal{B}$ and $\mathcal{B}'$ are ordered bases for a vector space V , then ${}_B[I]_{B'} = ({}_{B'}[I]_B)^{-1}$.
14. If $\mathcal{B}$ and $\mathcal{B}'$ are ordered bases for F^n , then ${}_{B'}[I]_{\mathcal{B}} = ({}_{B'}[I]_{{}_{\mathcal{E}_n}})({}_{\mathcal{E}_n}[I]_{\mathcal{B}})$.

Examples:

1. If $p(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0 \in F[x; n]$ with the standard ordered basis

$$\mathcal{B} = (1, x, x^2, \dots, x^n), \text{ then } [p(x)]_{\mathcal{B}} = \begin{bmatrix} a_0 \\ a_1 \\ \vdots \\ a_n \end{bmatrix}.$$

2. The set $\mathcal{B} = \left(\begin{bmatrix} 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ 3 \end{bmatrix} \right)$ forms an ordered basis for $\mathbb{R}^2$. If $\mathcal{E}_2$ is the standard ordered basis

for $\mathbb{R}^2$, then the change-of-basis matrix from $\mathcal{B}$ to $\mathcal{E}_2$ is ${}_{\mathcal{E}_2}[T]_{\mathcal{B}} = \begin{bmatrix} 1 & 0 \\ -1 & 3 \end{bmatrix}$, and $({}_{\mathcal{E}_2}[T]_{\mathcal{B}})^{-1} =$

$$\begin{bmatrix} 1 & 0 \\ \frac{1}{3} & \frac{1}{3} \end{bmatrix}. \text{ So for } \mathbf{v} = \begin{bmatrix} 3 \\ 1 \end{bmatrix} \text{ in the standard ordered basis, we find that } [\mathbf{v}]_{\mathcal{B}} = ({}_{\mathcal{E}_2}[T]_{\mathcal{B}})^{-1} \mathbf{v} = \begin{bmatrix} 3 \\ \frac{4}{3} \end{bmatrix}.$$

To check this, we can easily see that $\mathbf{v} = \begin{bmatrix} 3 \\ 1 \end{bmatrix} = 3 \begin{bmatrix} 1 \\ -1 \end{bmatrix} + \frac{4}{3} \begin{bmatrix} 0 \\ 3 \end{bmatrix}$.

3. The set $\mathcal{B}' = (1, 1 + x, 1 + x^2)$ is an ordered basis for $\mathbb{R}[x; 2]$, and using the standard ordered basis

$$\mathcal{B} = (1, x, x^2) \text{ for } \mathbb{R}[x; 2] \text{ we have } {}_B[P]_{B'} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}. \text{ So, } ({}_B[P]_{B'})^{-1} = \begin{bmatrix} 1 & -1 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\text{and } [5 - 2x + 3x^2]_{B'} = ({}_B[P]_{B'})^{-1} \begin{bmatrix} 5 \\ -2 \\ 3 \end{bmatrix} = \begin{bmatrix} 4 \\ -2 \\ 3 \end{bmatrix}. \text{ Of course, we can see } 5 - 2x + 3x^2 =$$

$$4(1) - 2(1 + x) + 3(1 + x^2).$$

4. If we want to change from the ordered basis $\mathcal{B}_1 = \left(\begin{bmatrix} 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ 3 \end{bmatrix} \right)$ in $\mathbb{R}^2$ to the ordered basis $\mathcal{B}_2 =$

$$\left(\begin{bmatrix} 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 5 \\ 0 \end{bmatrix} \right), \text{ then the resulting change-of-basis matrix is } {}_{B_2}[T]_{B_1} = ({}_{\mathcal{E}_2}[T]_{B_2})^{-1} ({}_{\mathcal{E}_2}[T]_{B_1}) =$$

$$\begin{bmatrix} 2 & 5 \\ 1 & 0 \end{bmatrix}^{-1} \begin{bmatrix} 1 & 0 \\ -1 & 3 \end{bmatrix} = \begin{bmatrix} -1 & 3 \\ \frac{3}{5} & -\frac{6}{5} \end{bmatrix}.$$

5. Let $S = \{5 - 2x + 3x^2, 3 - x + 2x^2, 8 + 3x\}$ in $\mathbb{R}[x; 2]$ with the standard ordered basis $\mathcal{B} = (1, x, x^2)$. The matrix $A = \begin{bmatrix} 5 & 3 & 8 \\ -2 & -1 & 3 \\ 3 & 2 & 0 \end{bmatrix}$ contains the $\mathcal{B}$ -coordinate vectors for the polynomials in S and it has row echelon form $\begin{bmatrix} 5 & 3 & 8 \\ 0 & 1 & 31 \\ 0 & 0 & 1 \end{bmatrix}$. Since this row echelon form shows that A is nonsingular, we know by Fact 8 above that S is a basis for $\mathbb{R}[x; 2]$.

2.7 Idempotence and Nilpotence

Definitions:

A is an **idempotent** if $A^2 = A$.

A is **nilpotent** if, for some $k \geq 0$, $A^k = 0$.

Facts: All of the following facts except those with a specific reference are immediate from the definitions.

1. Every idempotent except the identity matrix is singular.
2. Let $A \in F^{n \times n}$. The following statements are equivalent.
 - (a) A is an idempotent.
 - (b) $I - A$ is an idempotent.
 - (c) If $\mathbf{v} \in \text{range}(A)$, then $A\mathbf{v} = \mathbf{v}$.
 - (d) $F^n = \ker A \oplus \text{range } A$.
 - (e) [HJ85, p. 37 and p. 148] A is similar to $\begin{bmatrix} I_k & 0 \\ 0 & 0 \end{bmatrix}$, for some $k \leq n$.
3. If A_1 and A_2 are idempotents of the same size and commute, then $A_1 A_2$ is an idempotent.
4. If A_1 and A_2 are idempotents of the same size and $A_1 A_2 = A_2 A_1 = 0$, then $A_1 + A_2$ is an idempotent.
5. If $A \in F^{n \times n}$ is nilpotent, then $A^n = 0$.
6. If A is nilpotent and B is of the same size and commutes with A , then AB is nilpotent.
7. If A_1 and A_2 are nilpotent matrices of the same size and $A_1 A_2 = A_2 A_1 = 0$, then $A_1 + A_2$ is nilpotent.

Examples:

1. $\begin{bmatrix} -8 & 12 \\ -6 & 9 \end{bmatrix}$ is an idempotent. $\begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix}$ is nilpotent.

References

- [Lay03] D. C. Lay. *Linear Algebra and Its Applications*, 3rd ed. Addison-Wesley, Reading, MA, 2003.
- [HK71] K. H. Hoffman and R. Kunze. *Linear Algebra*, 2nd ed. Prentice-Hall, Upper Saddle River, NJ, 1971.
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3

Linear Transformations

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3.1 Basic Concepts

Let V, W be vector spaces over a field F .

Definitions:

A **linear transformation** (or **linear mapping**) is a mapping $T: V \rightarrow W$ such that, for each $\mathbf{u}, \mathbf{v} \in V$, and for each $c \in F$, $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$, and $T(c\mathbf{u}) = cT(\mathbf{u})$.

V is called the **domain** of the linear transformation $T: V \rightarrow W$.

W is called the **codomain** of the linear transformation $T: V \rightarrow W$.

The **identity** transformation $I_V: V \rightarrow V$ is defined by $I_V(\mathbf{v}) = \mathbf{v}$ for each $\mathbf{v} \in V$. I_V is also denoted by I .

The **zero** transformation $0: V \rightarrow W$ is defined by $0(\mathbf{v}) = \mathbf{0}_W$ for each $\mathbf{v} \in V$.

A **linear operator** is a linear transformation $T: V \rightarrow V$.

Facts:

Let $T: V \rightarrow W$ be a linear transformation. The following facts can be found in almost any elementary linear algebra text, including [Lan70, IV§1], [Sta69, §3.1], [Goo03, Chapter 4], and [Lay03, §1.8].

1. $T(\sum_1^n a_i \mathbf{v}_i) = \sum_1^n a_i T(\mathbf{v}_i)$, for any $a_i \in F$, $\mathbf{v}_i \in V$, $i = 1, \dots, n$.
2. $T(\mathbf{0}_V) = \mathbf{0}_W$.
3. $T(-\mathbf{v}) = -T(\mathbf{v})$, for each $\mathbf{v} \in V$.
4. The identity transformation is a linear transformation.
5. The zero transformation is a linear transformation.

6. If $\mathcal{B} = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ is a basis for V , and $\mathbf{w}_1, \dots, \mathbf{w}_n \in W$, then there exists a unique $T: V \rightarrow W$ such that $T(\mathbf{v}_i) = \mathbf{w}_i$ for each i .

Examples:

Examples 1 to 9 are linear transformations.

1. $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ where $T \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{bmatrix} x + y \\ 2x - z \end{bmatrix}$.
2. $T: V \rightarrow V$, defined by $T(\mathbf{v}) = -\mathbf{v}$ for each $\mathbf{v} \in V$.
3. If $A \in F^{m \times n}$, $T: F^n \rightarrow F^m$, where $T(\mathbf{v}) = A\mathbf{v}$.
4. $T: F^{m \times n} \rightarrow F$, where $T(A) = \text{tr} A$.
5. Let $\mathcal{C}([0, 1])$ be the vector space of all continuous functions on $[0, 1]$ into $\mathbb{R}$, and let $T: \mathcal{C}([0, 1]) \rightarrow \mathbb{R}$ be defined by $T(f) = \int_0^1 f(t) dt$.
6. Let V be the vector space of all functions $f: \mathbb{R} \rightarrow \mathbb{R}$ that have derivatives of all orders, and $D: V \rightarrow V$ be defined by $D(f) = f'$.
7. The transformation, which rotates every vector in the plane $\mathbb{R}^2$ through an angle θ .
8. The projection T onto the xy -plane of $\mathbb{R}^3$, i.e., $T \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{bmatrix} x \\ y \\ 0 \end{bmatrix}$.
9. $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$, where $T(\mathbf{v}) = \mathbf{b} \times \mathbf{v}$, for some $\mathbf{b} \in \mathbb{R}^3$.

Examples 10 and 11 are not linear transformations.

10. $f: \mathbb{R}^2 \rightarrow \mathbb{R}^2$, where $f \begin{pmatrix} x \\ y \end{pmatrix} = \begin{bmatrix} y + 1 \\ x - y - 2 \end{bmatrix}$ is not a linear transformation because $f(\mathbf{0}) \neq \mathbf{0}$.
11. $f: \mathbb{R}^2 \rightarrow \mathbb{R}$, where $f \begin{pmatrix} x \\ y \end{pmatrix} = x^2$ is not a linear transformation because $f \left(2 \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right) = 4 \neq 2 = 2f \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} \right)$.

3.2 The Spaces $L(V, W)$ and $L(V, V)$

Let V, W be vector spaces over F .

Definitions:

$L(V, W)$ denotes the set of all linear transformations of V into W .

For each $T_1, T_2 \in L(V, W)$ the **sum** $T_1 + T_2$ is defined by $(T_1 + T_2)(\mathbf{v}) = T_1(\mathbf{v}) + T_2(\mathbf{v})$.

For each $c \in F$, $T \in L(V, W)$ the **scalar multiple** cT is defined by $(cT)(\mathbf{v}) = cT(\mathbf{v})$.

For each $T_1, T_2 \in L(V, V)$ the **product** $T_1 T_2$ is the composite mapping defined by $(T_1 T_2)(\mathbf{v}) = T_1(T_2(\mathbf{v}))$.

$T_1, T_2 \in L(V, V)$ **commute** if $T_1 T_2 = T_2 T_1$.

$T \in L(V, V)$ is a **scalar transformation** if, for some $c \in F$, $T(\mathbf{v}) = c\mathbf{v}$ for each $\mathbf{v} \in V$.

Facts:

Let $T, T_1, T_2 \in L(V, W)$. The following facts can be found in almost any elementary linear algebra text, including [Fin60, §3.2], [Lan70, IV §4], [Sta69, §3.6], [SW68, §4.3], and [Goo03, Chap. 4].

1. $T_1 + T_2 \in L(V, W)$.
2. $cT \in L(V, W)$.

3. If $T_1, T_2 \in L(V, V)$, then $T_1 T_2 \in L(V, V)$.
4. $L(V, W)$, with sum and scalar multiplication, is a vector space over F .
5. $L(V, V)$, with sum, scalar multiplication, and composition, is a linear algebra over F .
6. Let $\dim V = n$ and $\dim W = m$. Then $\dim L(V, W) = mn$.
7. If $\dim V > 1$, then there exist $T_1, T_2 \in L(V, V)$, which do not commute.
8. $T_0 \in L(V, V)$ commutes with all $T \in L(V, V)$ if and only if T_0 is a scalar transformation.

Examples:

1. For each $j = 1, \dots, n$ let $T_j \in L(F^n, F^n)$ be defined by $T_j(\mathbf{x}) = x_j \mathbf{e}_j$. Then $\sum_{i=1}^n T_i$ is the identity transformation in V .
2. Let T_1 and T_2 be the transformations that rotates every vector in $\mathbb{R}^2$ through an angle θ_1 and θ_2 respectively. Then $T_1 T_2$ is the rotation through the angle $\theta_1 + \theta_2$.
3. Let T_1 be the rotation through an angle θ in $\mathbb{R}^2$ and let T_2 be the reflection on the horizontal axis, that is, $T_2(x, y) = (x, -y)$. Then T_1 and T_2 do not commute.

3.3 Matrix of a Linear Transformation

Let V, W be nonzero finite dimensional vector spaces over F .

Definitions:

The **linear transformation associated to a matrix** $A \in F^{m \times n}$ is $T_A: F^n \rightarrow F^m$ defined by $T_A(\mathbf{v}) = A\mathbf{v}$.

The **matrix associated to a linear transformation** $T \in L(V, W)$ and relative to the ordered bases $\mathcal{B} = (\mathbf{b}_1, \dots, \mathbf{b}_n)$ of V , and $\mathcal{C}$ of W , is the matrix ${}_C[T]_B = [[T(\mathbf{b}_1)]_C \cdots [T(\mathbf{b}_n)]_C]$.

If $T \in L(F^n, F^m)$, then the **standard matrix** of T is $[T] = {}_{\mathcal{E}_m}[T]_{\mathcal{E}_n}$, where $\mathcal{E}_n$ is the standard basis for F^n .

Note: If $V = W$ and $\mathcal{B} = \mathcal{C}$, the matrix ${}_B[T]_B$ will be denoted by $[T]_B$.

If $T \in L(V, V)$ and $\mathcal{B}$ is an ordered basis for V , then the **trace** of T is $\text{tr } T = \text{tr } [T]_B$.

Facts:

Let $\mathcal{B}$ and $\mathcal{C}$ be ordered bases V and W , respectively. The following facts can be found in almost any elementary linear algebra text, including [Lan70, V §2], [Sta69, §3.4–3.6], [SW68, §4.3], and [Goo03, Chap. 4].

1. The trace of $T \in L(V, V)$ is independent of the ordered basis of V used to define it.
2. For $A, B \in F^{m \times n}$, $T_A = T_B$ if and only if $A = B$.
3. For any $T_1, T_2 \in L(V, W)$, ${}_C[T_1]_B = {}_C[T_2]_B$ if and only if $T_1 = T_2$.
4. If $T \in L(F^n, F^m)$, then $[T] = [T(\mathbf{e}_1) \cdots T(\mathbf{e}_n)]$.
5. The change-of-basis matrix from basis $\mathcal{B}$ to $\mathcal{C}$, ${}_C[I]_B$, as defined in Chapter 2.6, is the same matrix as the matrix of the identity transformation with respect to $\mathcal{B}$ and $\mathcal{C}$.
6. Let $A \in F^{m \times n}$ and let T_A be the linear transformation associated to A . Then $[T_A] = A$.
7. If $T \in L(F^n, F^m)$, then ${}_{[T]}[T] = T$.
8. For any $T_1, T_2 \in L(V, W)$, ${}_C[T_1 + T_2]_B = {}_C[T_1]_B + {}_C[T_2]_B$.
9. For any $T \in L(V, W)$, and $c \in F$, ${}_C[cT]_B = c {}_C[T]_B$.
10. For any $T_1, T_2 \in L(V, V)$, $[T_1 T_2]_B = [T_1]_B [T_2]_B$.
11. If $T \in L(V, W)$, then, for each $\mathbf{v} \in V$, $[T(\mathbf{v})]_C = {}_C[T]_B [\mathbf{v}]_B$. Furthermore ${}_C[T]_B$ is the only matrix A such that, for each $\mathbf{v} \in V$, $[T(\mathbf{v})]_C = A[\mathbf{v}]_B$.

Examples:

1. Let T be the projection of $\mathbb{R}^3$ onto the xy -plane of $\mathbb{R}^3$. Then

$$[T] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

2. Let T be the identity in F^n . Then $[T]_{\mathcal{B}} = I_n$.
3. Let T be the rotation by θ in $\mathbb{R}^2$. Then

$$[T] = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}.$$

4. Let $D: \mathbb{R}[x; n] \rightarrow \mathbb{R}[x; n-1]$ be the derivative transformation, and let $\mathcal{B} = \{1, x, \dots, x^n\}$, $\mathcal{C} = \{1, x, \dots, x^{n-1}\}$. Then

$${}_c[T]_{\mathcal{B}} = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 2 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & n-1 \end{bmatrix}.$$

3.4 Change of Basis and Similarity

Let V, W be nonzero finite dimensional vector spaces over F .

Facts:

The following facts can be found in [Gan60, III §5–6] and [Goo03, Chap. 4].

1. Let $T \in L(V, W)$ and let $\mathcal{B}, \mathcal{B}'$ be bases of $V, \mathcal{C}, \mathcal{C}'$ be bases of W . Then

$${}_{\mathcal{C}'}[T]_{\mathcal{B}'} = {}_{\mathcal{C}'}[I]_{\mathcal{C}} {}_{\mathcal{C}}[T]_{\mathcal{B}} [I]_{\mathcal{B}'}.$$

2. Two $m \times n$ matrices are equivalent if and only if they represent the same linear transformation $T \in L(V, W)$, but possibly in different bases, as in Fact 1.
3. Any $m \times n$ matrix A of rank r is equivalent to the $m \times n$ matrix

$$\tilde{I}_r = \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}.$$

4. Two $m \times n$ matrices are equivalent if and only if they have the same rank.
5. Two $n \times n$ matrices are similar if and only if they represent the same linear transformation $T \in L(V, V)$, but possibly in different bases, i.e., if A_1 is similar to A_2 , then there is $T \in L(V, V)$ and ordered bases $\mathcal{B}_1, \mathcal{B}_2$ of V such that $A_i = [T]_{\mathcal{B}_i}$ and conversely.

Examples:

1. Let T be the projection on the x -axis of $\mathbb{R}^2$, i.e., $T(x, y) = (x, 0)$. If $\mathcal{B} = \{\mathbf{e}_1, \mathbf{e}_2\}$ and $\mathcal{C} = \{\mathbf{e}_1 + \mathbf{e}_2, \mathbf{e}_1 - \mathbf{e}_2\}$, then $[T]_{\mathcal{B}} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$, $[T]_{\mathcal{C}} = \begin{bmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{bmatrix}$, and $[T]_{\mathcal{C}} = Q^{-1}[T]_{\mathcal{B}}Q$ with $Q = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$.

3.5 Kernel and Range

Let V, W be vector spaces over F and let $T \in L(V, W)$.

Definitions:

T is **one-to-one** (or **injective**) if $\mathbf{v}_1 \neq \mathbf{v}_2$ implies $T(\mathbf{v}_1) \neq T(\mathbf{v}_2)$.

The **kernel** (or **null space**) of T is the set $\ker T = \{\mathbf{v} \in V \mid T(\mathbf{v}) = \mathbf{0}\}$.

The **nullity** of T , denoted by $\text{null } T$, is the dimension of $\ker T$.

T is **onto** (or **surjective**) if, for each $\mathbf{w} \in W$, there exists $\mathbf{v} \in V$ such that $T(\mathbf{v}) = \mathbf{w}$.

The **range** (or **image**) of T is the set $\text{range } T = \{\mathbf{w} \in W \mid \exists \mathbf{v}, \mathbf{w} = T(\mathbf{v})\}$.

The **rank** of T , denoted by $\text{rank } T$, is the dimension of $\text{range } T$.

Facts:

The following facts can be found in [Fin60, §3.3], [Lan70, IV §3], [Sta69, §3.1–3.2], and [Goo03, Chap. 4].

1. $\ker T$ is a subspace of V .
2. The following statements are equivalent.
 - (a) T is one-to-one.
 - (b) $\ker T = \{\mathbf{0}\}$.
 - (c) Each linearly independent set is mapped to a linearly independent set.
 - (d) Each basis is mapped to a linearly independent set.
 - (e) Some basis is mapped to a linearly independent set.
3. $\text{range } T$ is a subspace of W .
4. $\text{rank } T = \text{rank}_C [T]_B$ for any finite nonempty ordered bases B, C .
5. For $A \in F^{m \times n}$, $\ker T_A = \ker A$ and $\text{range } T_A = \text{range } A$.
6. (Dimension Theorem) Let $T \in L(V, W)$ where V has finite dimension. Then $\text{null } T + \text{rank } T = \dim V$.
7. Let $T \in L(V, V)$, where V has finite dimension, then T is one-to-one if and only if T is onto.
8. Let $T(\mathbf{v}) = \mathbf{w}$. Then $\{\mathbf{u} \in V \mid T(\mathbf{u}) = \mathbf{w}\} = \mathbf{v} + \ker T$.
9. Let $V = \text{Span}\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$. Then $\text{range } T = \text{Span}\{T(\mathbf{v}_1), \dots, T(\mathbf{v}_n)\}$.
10. Let $T_1, T_2 \in L(V, V)$. Then $\ker T_1 T_2 \supseteq \ker T_2$ and $\text{range } T_1 T_2 \subseteq \text{range } T_1$.
11. Let $T \in L(V, V)$. Then

$$\{\mathbf{0}\} \subseteq \ker T \subseteq \ker T^2 \subseteq \dots \subseteq \ker T^k \subseteq \dots$$

$$V \supseteq \text{range } T \supseteq \text{range } T^2 \supseteq \dots \supseteq \text{range } T^k \supseteq \dots$$

Furthermore, if, for some k , $\text{range } T^{k+1} = \text{range } T^k$, then, for each $i \geq 1$, $\text{range } T^{k+i} = \text{range } T^k$.
 If, for some k , $\ker T^{k+1} = \ker T^k$, then, for each $i \geq 1$, $\ker T^{k+i} = \ker T^k$.

Examples:

1. Let T be the projection of $\mathbb{R}^3$ onto the xy -plane of $\mathbb{R}^3$. Then $\ker T = \{(0, 0, z) : z \in \mathbb{R}\}$; $\text{range } T = \{(x, y, 0) : x, y \in \mathbb{R}\}$; $\text{null } T = 1$; and $\text{rank } T = 2$.
2. Let T be the linear transformation in Example 1 of Section 3.1. Then $\ker T = \text{Span}\{[1 \ -1 \ 2]^T\}$, while $\text{range } T = \mathbb{R}^2$.
3. Let $D \in L(\mathbb{R}[x], \mathbb{R}[x])$ be the derivative transformation, then $\ker D$ consists of all constant polynomials, while $\text{range } D = \mathbb{R}[x]$. In particular, D is onto but is not one-to-one. Note that $\mathbb{R}[x]$ is not finite dimensional.

4. Let $T_1, T_2 \in L(F^{n \times n}, F^{n \times n})$ where $T_1(A) = \frac{1}{2}(A - A^T)$, $T_2(A) = \frac{1}{2}(A + A^T)$, then

$$\ker T_1 = \text{range } T_2 = \{n \times n \text{ symmetric matrices}\};$$

$$\ker T_2 = \text{range } T_1 = \{n \times n \text{ skew-symmetric matrices}\};$$

$$\text{null } T_1 = \text{rank } T_2 = \frac{n(n+1)}{2}; \quad \text{null } T_2 = \text{rank } T_1 = \frac{n(n-1)}{2}.$$

5. Let $T(\mathbf{v}) = \mathbf{b} \times \mathbf{v}$ as in Example 9 of Section 3.1. Then $\ker T = \text{Span}\{\mathbf{b}\}$.

3.6 Invariant Subspaces and Projections

Let V be a vector space over F , and let $V = V_1 \oplus V_2$ for some V_1, V_2 subspaces of V . For each $\mathbf{v} \in V$, let $\mathbf{v}_i \in V_i$ denote the (unique) vector such that $\mathbf{v} = \mathbf{v}_1 + \mathbf{v}_2$ (see Section 2.3). Finally, let $T \in L(V, V)$.

Definitions:

For $i, j \in \{1, 2\}$, $i \neq j$, the **projection** onto V_i along V_j is the operator $\text{proj}_{V_i, V_j} : V \rightarrow V$ defined by $\text{proj}_{V_i, V_j}(\mathbf{v}) = \mathbf{v}_i$ for each $\mathbf{v} \in V$ (see also Chapter 5).

The **complementary projection** of the projection proj_{V_i, V_j} is the projection proj_{V_j, V_i} .

T is an **idempotent** if $T^2 = T$.

A subspace V_0 of V is **invariant** under T or **T -invariant** if $T(V_0) \subseteq V_0$.

The **fixed space** of T is $\text{fix } T = \{\mathbf{v} \in V \mid T(\mathbf{v}) = \mathbf{v}\}$.

T is **nilpotent** if, for some $k \geq 0$, $T^k = 0$.

Facts:

The following facts can be found in [Mal63, §43–44].

1. $\text{proj}_{V_i, V_j} \in L(V, V)$.
2. $\text{proj}_{V_1, V_2} + \text{proj}_{V_2, V_1} = I$, the identity linear operator in V .
3. $\text{range}(\text{proj}_{V_i, V_j}) = \ker(\text{proj}_{V_j, V_i}) = V_i$.
4. Sum and intersection of invariant subspaces are invariant subspaces.
5. If V has a nonzero subspace different from V that is invariant under T , then there exists a suitable ordered basis $\mathcal{B}$ of V such that $[T]_{\mathcal{B}} = \begin{bmatrix} A_{11} & A_{12} \\ 0 & A_{22} \end{bmatrix}$. Conversely, if $[T]_{\mathcal{B}} = \begin{bmatrix} A_{11} & A_{12} \\ 0 & A_{22} \end{bmatrix}$, where A_{11} is an m -by- m block, then the subspace spanned by the first m vectors in $\mathcal{B}$ is a T -invariant subspace.
6. Let T have two nonzero finite dimensional invariant subspaces V_1 and V_2 , with ordered bases $\mathcal{B}_1$ and $\mathcal{B}_2$, respectively, such that $V_1 \oplus V_2 = V$. Let $T_1 \in L(V_1, V_1)$, $T_2 \in L(V_2, V_2)$ be the restrictions of T on V_1 and V_2 , respectively, and let $\mathcal{B} = \mathcal{B}_1 \cup \mathcal{B}_2$. Then $[T]_{\mathcal{B}} = [T_1]_{\mathcal{B}_1} \oplus [T_2]_{\mathcal{B}_2}$.

The following facts can be found in [Hoh64, §6.15; §6.20].

7. Every idempotent except the identity is singular.
8. The statements 8a through 8e are equivalent. If V is finite dimensional, statement 8f is also equivalent to these statements.
 - (a) T is an idempotent.
 - (b) $I - T$ is an idempotent.
 - (c) $\text{fix } T = \text{range } T$.
 - (d) $V = \ker T \oplus \text{fix } T$.

- (e) T is the projection onto V_1 along V_2 for some V_1, V_2 , with $V = V_1 \oplus V_2$.
- (f) There exists a basis $\mathcal{B}$ of V such that $[T]_{\mathcal{B}} = \begin{bmatrix} I & 0 \\ 0 & 0 \end{bmatrix}$.
9. If T_1 and T_2 are idempotents on V and commute, then $T_1 T_2$ is an idempotent.
10. If T_1 and T_2 are idempotents on V and $T_1 T_2 = T_2 T_1 = 0$, then $T_1 + T_2$ is an idempotent.
11. If $\dim V = n$ and $T \in L(V, V)$ is nilpotent, then $T^n = 0$.

Examples:

- Example 8 of Section 3.1, $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$, where $T \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x \\ y \\ 0 \end{pmatrix}$ is the projection onto $\text{Span}\{\mathbf{e}_1, \mathbf{e}_2\}$ along $\text{Span}\{\mathbf{e}_3\}$.
- The zero subspace is T -invariant for any T .
- T_1 and T_2 , defined in Example 4 of Section 3.5, are the projection of $F^{n \times n}$ onto the subspace of n -by- n symmetric matrices along the subspace of n -by- n skew-symmetric matrices, and the projection of $F^{n \times n}$ onto the skew-symmetric matrices along the symmetric matrices, respectively.
- Let T be a nilpotent linear transformation on V . Let $T^p = 0$ and $T^{p-1}(\mathbf{v}) \neq 0$. Then $S = \text{Span}\{\mathbf{v}, T(\mathbf{v}), T^2(\mathbf{v}), \dots, T^{p-1}(\mathbf{v})\}$ is a T -invariant subspace.

3.7 Isomorphism and Nonsingularity Characterization

Let U, V, W be vector spaces over F and let $T \in L(V, W)$.

Definitions:

T is **invertible** (or an **isomorphism**) if there exists a function $S: W \rightarrow V$ such that $ST = I_V$ and $TS = I_W$. S is called the **inverse** of T and is denoted by T^{-1} .

V and W are **isomorphic** if there exists an isomorphism of V onto W .

T is **nonsingular** if $\ker T = \{\mathbf{0}\}$; otherwise T is **singular**.

Facts:

The following facts can be found in [Fin60, §3.4], [Hoh64, §6.11], and [Lan70, IV §4]:

- The inverse is unique.
- T^{-1} is a linear transformation, invertible, and $(T^{-1})^{-1} = T$.
- If $T_1 \in L(V, W)$ and $T_2 \in L(U, V)$, then $T_1 T_2$ is invertible if and only if T_1 and T_2 are invertible.
- If $T_1 \in L(V, W)$ and $T_2 \in L(U, V)$, then $(T_1 T_2)^{-1} = T_2^{-1} T_1^{-1}$.
- Let $T \in L(V, W)$, and let $\dim V = \dim W = n$. The following statements are equivalent:
 - T is invertible.
 - T is nonsingular.
 - T is one-to-one.
 - $\ker T = \{\mathbf{0}\}$.
 - $\text{null } T = 0$.
 - T is onto.
 - $\text{range } T = W$.

- (h) $\text{rank } T = n$.
- (i) T maps some bases of V to bases of W .
- 6. If V and W are isomorphic, then $\dim V = \dim W$.
- 7. If $\dim V = n > 0$, then V is isomorphic to F^n through φ defined by $\varphi(\mathbf{v}) = [\mathbf{v}]_{\mathcal{B}}$ for any ordered basis $\mathcal{B}$ of V .
- 8. Let $\dim V = n > 0$, $\dim W = m > 0$, and let $\mathcal{B}$ and $\mathcal{C}$ be ordered bases of V and W , respectively. Then $L(V, W)$ and $F^{m \times n}$ are isomorphic through φ defined by $\varphi(T) = {}_{\mathcal{C}}[T]_{\mathcal{B}}$.

Examples:

1. $V = F[x; n]$ and $W = F^{n+1}$ are isomorphic through $T \in L(V, W)$ defined by $T(\sum_0^n a_i x^i) = [a_0 \dots a_n]^T$.
2. If V is an infinite dimensional vector space, a nonsingular linear operator $T \in L(V, V)$ need not be invertible. For example, let $T \in L(\mathbb{R}[x], \mathbb{R}[x])$ be defined by $T(p(x)) = xp(x)$. Then T is nonsingular but not invertible since T is not onto. For matrices, nonsingular and invertible are equivalent, since an $n \times n$ matrix over F is an operator on the finite dimensional vector F^n .

3.8 Linear Functionals and Annihilator

Let V, W be vector spaces over F .

Definitions:

A **linear functional** (or **linear form**) on V is a linear transformation from V to F .

The **dual space** of V is the vector space $V^* = L(V, F)$ of all linear functionals on V .

If V is nonzero and finite dimensional, the **dual basis** of a basis $\mathcal{B} = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ of V is the set $\mathcal{B}^* = \{f_1, \dots, f_n\} \subseteq V^*$, such that $f_i(\mathbf{v}_j) = \delta_{ij}$ for each i, j .

The **bidual space** is the vector space $V^{**} = (V^*)^* = L(V^*, F)$.

The **annihilator** of a set $S \subseteq V$ is $S^a = \{f \in V^* \mid f(\mathbf{v}) = 0, \forall \mathbf{v} \in S\}$.

The **transpose** of $T \in L(V, W)$ is the mapping $T^T \in L(W^*, V^*)$ defined by setting, for each $g \in W^*$,

$$\begin{aligned} T^T(g) : V &\rightarrow F \\ \mathbf{v} &\mapsto g(T(\mathbf{v})). \end{aligned}$$

Facts:

The following facts can be found in [Hoh64, §6.19] and [SW68, §4.4].

1. For each $\mathbf{v} \in V$, $\mathbf{v} \neq 0$, there exists $f \in V^*$ such that $f(\mathbf{v}) \neq 0$.
2. For each $\mathbf{v} \in V$ define $h_{\mathbf{v}} \in L(V^*, F)$ by setting $h_{\mathbf{v}}(f) = f(\mathbf{v})$. Then the mapping

$$\begin{aligned} \varphi : V &\rightarrow V^{**} \\ \mathbf{v} &\mapsto h_{\mathbf{v}} \end{aligned}$$

is a one-to-one linear transformation. If V is finite dimensional, φ is an isomorphism of V onto V^{**} .

3. S^a is a subspace of V^* .
4. $\{0\}^a = V^*$; $V^a = \{0\}$.
5. $S^a = (\text{Span}\{S\})^a$.

The following facts hold for finite dimensional vector spaces.

6. If V is nonzero, for each basis $\mathcal{B}$ of V , the dual basis exists, is uniquely determined, and is a basis for V^* .
7. $\dim V = \dim V^*$.
8. If V is nonzero, each basis of V^* is the dual basis of some basis of V .

9. Let $\mathcal{B}$ be a basis for the nonzero vector space V . For each $\mathbf{v} \in V$, $f \in V^*$, $f(\mathbf{v}) = [f]_{\mathcal{B}^*}^T [\mathbf{v}]_{\mathcal{B}}$.
10. If S is a subspace of V , then $\dim S + \dim S^a = \dim V$.
11. If S is a subspace of V , then, by identifying V and V^{**} , $S = (S^a)^a$.
12. Let S_1, S_2 be subspaces of V such that $S_1^a = S_2^a$. Then $S_1 = S_2$.
13. Any subspace of V^* is the annihilator of some subspace S of V .
14. Let S_1, S_2 be subspaces of V . Then $(S_1 \cap S_2)^a = S_1^a + S_2^a$ and $(S_1 + S_2)^a = S_1^a \cap S_2^a$.
15. $\ker T^T = (\text{range } T)^a$.
16. $\text{rank } T = \text{rank } T^T$.
17. If $\mathcal{B}$ and $\mathcal{C}$ are nonempty bases of V and W , respectively, then $_{\mathcal{B}^*}[T^T]_{\mathcal{C}^*} = ({}_c[T]_{\mathcal{B}})^T$.

Examples:

1. Let $V = \mathcal{C}[a, b]$ be the vector space of continuous functions $\varphi : [a, b] \rightarrow \mathbb{R}$, and let $c \in [a, b]$. Then $f(\varphi) = \varphi(c)$ is a linear functional on V .
2. Let $V = \mathcal{C}[a, b]$, $\psi \in V$, and $f(\varphi) = \int_a^b \varphi(t)\psi(t)dt$. Then f is a linear functional.
3. The trace is a linear functional on $F^{n \times n}$.
4. let $V = F^{m \times n}$. $\mathcal{B} = \{E_{ij} : 1 \leq i \leq m, 1 \leq j \leq n\}$ is a basis for V . The dual basis $\mathcal{B}^*$ consists of the linear functionals f_{ij} , $1 \leq i \leq m, 1 \leq j \leq n$, defined by $f_{ij}(A) = a_{ij}$.

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4

Determinants and Eigenvalues

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4.1 Determinants

Definitions:

The **determinant**, $\det A$, of a matrix $A = [a_{ij}] \in F^{n \times n}$ is an element in F defined inductively:

- $\det [a] = a$.
- For $i, j \in \{1, 2, \dots, n\}$, the **ij th minor** of A corresponding to a_{ij} is defined by $m_{ij} = \det A(\{i\}, \{j\})$.
- The **ij th cofactor** of a_{ij} is $c_{ij} = (-1)^{i+j} m_{ij}$.
- $\det A = \sum_{j=1}^n (-1)^{i+j} a_{ij} m_{ij} = \sum_{j=1}^n a_{ij} c_{ij}$ for $i \in \{1, 2, \dots, n\}$.

This method of computing the determinant of a matrix is called **Laplace expansion of the determinant by minors along the i th row**.

The **determinant of a linear operator** $T: V \rightarrow V$ on a finite dimensional vector space, V , is defined as $\det(T) = \det([T]_{\mathcal{B}})$, where $\mathcal{B}$ is a basis for V .

Facts:

All matrices are assumed to be in $F^{n \times n}$, unless otherwise stated. All the following facts except those with a specific reference can be found in [Lay03, pp. 185–213] or [Goo03, pp. 167–193].

1. $\det \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} = a_{11}a_{22} - a_{12}a_{21}$.
2. $\det A = \det \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = a_{11}a_{22}a_{33} + a_{21}a_{13}a_{32} + a_{31}a_{12}a_{23} - a_{31}a_{13}a_{22} - a_{21}a_{12}a_{33} - a_{11}a_{32}a_{23}$.
3. The determinant is independent of the row i used to evaluate it.
4. (Expansion of the determinant by minors along the j th column) Let $j \in \{1, 2, \dots, n\}$. Then $\det A = \sum_{i=1}^n (-1)^{i+j} a_{ij} m_{ij} = \sum_{i=1}^n a_{ij} c_{ij}$.
5. $\det I_n = 1$.
6. If A is a triangular matrix, then $\det A = a_{11}a_{22} \cdots a_{nn}$.

7. If B is a matrix obtained from A by interchanging two rows (or columns), then $\det B = -\det A$.
8. If B is a matrix obtained from A by multiplying one row (or column) by a nonzero constant r , then $\det B = r \det A$.
9. If B is a matrix obtained from A by adding to a row (or column) a multiple of another row (or column), then $\det B = \det A$.
10. If A , B , and C differ only in the r th row (or column), and the r th row (or column) of C is the sum of the r th rows (or columns) of A and B , then $\det C = \det A + \det B$.
11. If A is a matrix with a row (or column) of zeros, then $\det A = 0$.
12. If A is a matrix with two identical rows (or columns), then $\det A = 0$.
13. Let B be a row echelon form of A obtained by Gaussian elimination, using k row interchange operations and adding multiples of one row to another (see Algorithm 1 in Section 1.3). Then $\det A = (-1)^k \det B = (-1)^k b_{11} b_{22} \cdots b_{nn}$.
14. $\det A^T = \det A$.
15. If $A \in \mathbb{C}^{n \times n}$, then $\det A^* = \overline{\det A}$.
16. $\det AB = \det A \det B$.
17. If $c \in F$, then $\det(cA) = c^n \det A$.
18. A is nonsingular, that is A^{-1} exists, if and only if $\det A \neq 0$.
19. If A is nonsingular, then $\det(A^{-1}) = \frac{1}{\det A}$.
20. If S is nonsingular, then $\det(S^{-1}AS) = \det A$.
21. [HJ85] $\det A = \sum_{\sigma} \operatorname{sgn} \sigma a_{1\sigma(1)} a_{2\sigma(2)} \cdots a_{n\sigma(n)}$, where summation is over the $n!$ permutations, σ , of the n indices $\{1, 2, \dots, n\}$. The weight “ $\operatorname{sgn} \sigma$ ” is 1 when σ is even and -1 when σ is odd. (See Preliminaries for more information on permutations.)
22. If $\mathbf{x}, \mathbf{y} \in F^n$, then $\det(I + \mathbf{xy}^T) = 1 + \mathbf{y}^T \mathbf{x}$.
23. [FIS89] Let T be a linear operator on a finite dimensional vector space V . Let $\mathcal{B}$ and $\mathcal{B}'$ be bases for V . Then $\det(T) = \det([T]_{\mathcal{B}}) = \det([T]_{\mathcal{B}'})$.
24. [FIS89] Let T be a linear operator on a finite dimensional vector space V . Then T is invertible if and only if $\det(T) \neq 0$.
25. [FIS89] Let T be an invertible linear operator on a finite dimensional vector space V . Then $\det(T^{-1}) = \frac{1}{\det(T)}$.
26. [FIS89] Let T and U be linear operators on a finite dimensional vector space V . Then $\det(TU) = \det(T) \cdot \det(U)$.

Examples:

1. Let $A = \begin{bmatrix} 3 & -2 & 4 \\ 2 & 5 & -6 \\ -3 & 1 & 5 \end{bmatrix}$. Expanding the determinant of A along the second column: $\det A = 2 \cdot \det \begin{bmatrix} 2 & -6 \\ -3 & 5 \end{bmatrix} + 5 \cdot \det \begin{bmatrix} 3 & 4 \\ -3 & 5 \end{bmatrix} - \det \begin{bmatrix} 3 & 4 \\ 2 & -6 \end{bmatrix} = 2 \cdot (-8) + 5 \cdot 27 + 26 = 145$.
2. Let $A = \begin{bmatrix} -1 & 3 & -2 & 4 \\ 2 & 5 & 8 & 1 \\ 7 & -4 & 0 & -6 \\ 0 & 3 & 1 & 5 \end{bmatrix}$. Expanding the determinant of A along the third row: $\det A = 7 \cdot \det \begin{bmatrix} 3 & -2 & 4 \\ 5 & 8 & 1 \\ 0 & 3 & 1 \end{bmatrix} + 4 \cdot \det \begin{bmatrix} -1 & -2 & 4 \\ 2 & 8 & 1 \\ 0 & 1 & 5 \end{bmatrix} + 6 \cdot \det \begin{bmatrix} -1 & 3 & -2 \\ 2 & 5 & 8 \\ 0 & 3 & 1 \end{bmatrix} = 557$.
3. Let $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $T \left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \right) = \begin{bmatrix} 2x_1 - 3x_2 \\ x_1 + 6x_2 \end{bmatrix}$. With $\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$, then $\det([T]_{\mathcal{B}}) = \det \begin{bmatrix} 2 & -3 \\ 1 & 6 \end{bmatrix} = 15$. Now let $\mathcal{B}' = \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\}$. Then $\det([T]_{\mathcal{B}'}) = \det \begin{bmatrix} 7 & 1 \\ -8 & 1 \end{bmatrix} = 15$.

Applications:

1. (Cramer's Rule) If $A \in F^{n \times n}$ is nonsingular, then the equation $A\mathbf{x} = \mathbf{b}$, where $\mathbf{x}, \mathbf{b} \in F^n$, has the

unique solution $\mathbf{s} = \begin{bmatrix} s_1 \\ s_2 \\ \vdots \\ s_n \end{bmatrix}$, where $s_i = \frac{\det A_i}{\det A}$ and A_i is the matrix obtained from A by replacing the i th column with $\mathbf{b}$.

2. [Mey00, p. 486] (Vandermonde Determinant) $\det \begin{bmatrix} 1 & 1 & \cdots & 1 \\ x_1 & x_2 & \cdots & x_n \\ x_1^2 & x_2^2 & \cdots & x_n^2 \\ \vdots & \vdots & \ddots & \vdots \\ x_1^{n-1} & x_2^{n-1} & \cdots & x_n^{n-1} \end{bmatrix} = \prod_{1 \leq i < j \leq n} (x_i - x_j).$

3. [FB90, pp. 220–235] (Volume) Let $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n$ be linearly independent vectors in $\mathbb{R}^m$. The volume, V , of the n -dimensional solid in $\mathbb{R}^m$, defined by $S = \{\sum_{i=1}^n t_i \mathbf{a}_i, 0 \leq t_i \leq 1, i = 1, 2, \dots, n\}$, is given by $V = \sqrt{\det(A^T A)}$, where A is the matrix whose i th column is the vector $\mathbf{a}_i$.

Let $m \geq n$ and $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation whose standard matrix representation is the $m \times n$ matrix A . Let S be a region in $\mathbb{R}^n$ of volume V_S . Then the volume of the image of S under the transformation T is $V_{T(S)} = \sqrt{\det(A^T A)} \cdot V_S$.

4. [Uhl02, pp. 247–248] (Wronskian) Let $f_1, f_2, \dots, f_n$ be $n - 1$ times differentiable functions of the real variable x . The determinant

$$W(f_1, f_2, \dots, f_n)(x) = \det \begin{bmatrix} f_1(x) & f_2(x) & \cdots & f_n(x) \\ f_1'(x) & f_2'(x) & \cdots & f_n'(x) \\ \vdots & \vdots & \ddots & \vdots \\ f_1^{(n-1)}(x) & f_2^{(n-1)}(x) & \cdots & f_n^{(n-1)}(x) \end{bmatrix}$$

is called the **Wronskian** of $f_1, f_2, \dots, f_n$. If $W(f_1, f_2, \dots, f_n)(x) \neq 0$ for some $x \in \mathbb{R}$, then the functions $f_1, f_2, \dots, f_n$ are linearly independent.

4.2 Determinants: Advanced Results

Definitions:

A **principal minor** is the determinant of a principal submatrix. (See Section 1.2.)

A **leading principal minor** is the determinant of a leading principal submatrix.

The sum of all the $k \times k$ principal minors of A is denoted $S_k(A)$.

The **k th compound matrix** of $A \in F^{m \times n}$ is the $\binom{m}{k} \times \binom{n}{k}$ matrix $C_k(A)$ whose entries are the $k \times k$ minors of A , usually in lexicographical order.

The **adjugate** of $A \in F^{n \times n}$ is the matrix $\text{adj } A = [c_{ji}] = [c_{ij}]^T$, where c_{ij} is the ij th-cofactor.

The **k th adjugate** of $A \in F^{n \times n}$ is the $\binom{n}{k} \times \binom{n}{k}$ matrix $\text{adj}^{(k)} A$, whose a_{ji} entry is the cofactor, in A , of the $(n - k)$ th minor, of A , in the ij th position of the compound.

Let $\alpha \subseteq \{1, 2, \dots, n\}$ and $A \in F^{n \times n}$ with $A[\alpha]$ nonsingular. The matrix

$$A/A[\alpha] = A[\alpha^c] - A[\alpha^c, \alpha]A[\alpha]^{-1}A[\alpha, \alpha^c]$$

is called the **Schur complement** of $A[\alpha]$.

Facts:

All matrices are assumed to be in $F^{n \times n}$, unless otherwise stated. All the following facts except those with a specific reference can be found in [Lay03, pp. 185–213] or [Goo03, pp. 167–193].

1. $A(\operatorname{adj} A) = (\operatorname{adj} A)A = (\det A)I_n$.
2. $\det(\operatorname{adj} A) = (\det A)^{n-1}$.
3. If $\det A \neq 0$, then $\operatorname{adj} A$ is nonsingular, and $(\operatorname{adj} A)^{-1} = (\det A)^{-1}A$.

4. [Ait56] (Method of Condensation) Let $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & a_{n3} & \cdots & a_{nn} \end{bmatrix}$, and assume without

loss of generality that $a_{11} \neq 0$, otherwise a nonzero element can be brought to the $(1, 1)$ position by interchanging two rows, which will change the sign of the determinant. Multiply all the rows of A except the first by a_{11} . For $i = 2, 3, \dots, n$, perform the row operations: replace row i with row $i - a_{i1}$.

$$\text{row 1. Thus } a_{11}^{n-1} \det A = \det \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ 0 & a_{11}a_{22} - a_{21}a_{12} & a_{11}a_{23} - a_{21}a_{13} & \cdots & a_{11}a_{2n} - a_{21}a_{1n} \\ 0 & a_{11}a_{32} - a_{31}a_{12} & a_{11}a_{33} - a_{31}a_{13} & \cdots & a_{11}a_{3n} - a_{31}a_{1n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & a_{11}a_{n2} - a_{n1}a_{12} & a_{11}a_{n3} - a_{n1}a_{13} & \cdots & a_{11}a_{nn} - a_{n1}a_{1n} \end{bmatrix}.$$

$$\text{So, } \det A = \frac{1}{a_{11}^{n-1}} \cdot \det \begin{bmatrix} \det \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} & \det \begin{bmatrix} a_{11} & a_{13} \\ a_{21} & a_{23} \end{bmatrix} & \cdots & \det \begin{bmatrix} a_{11} & a_{1n} \\ a_{21} & a_{2n} \end{bmatrix} \\ \det \begin{bmatrix} a_{11} & a_{12} \\ a_{31} & a_{32} \end{bmatrix} & \det \begin{bmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{bmatrix} & \cdots & \det \begin{bmatrix} a_{11} & a_{1n} \\ a_{31} & a_{3n} \end{bmatrix} \\ \vdots & \vdots & \ddots & \vdots \\ \det \begin{bmatrix} a_{11} & a_{12} \\ a_{n1} & a_{n2} \end{bmatrix} & \det \begin{bmatrix} a_{11} & a_{13} \\ a_{n1} & a_{n3} \end{bmatrix} & \cdots & \det \begin{bmatrix} a_{11} & a_{1n} \\ a_{n1} & a_{nn} \end{bmatrix} \end{bmatrix}.$$

5. [Ait56] $A^{(k)}(\operatorname{adj}^{(k)} A) = (\operatorname{adj}^{(k)} A)A^{(k)} = (\det A)I_n$.
6. [Ait56] $\det(A^{(k)}) = \det(A^r)$, where $r = \binom{n-1}{k-1}$.
7. [Ait56] $\det(A^{(n-k)}) = \det(\operatorname{adj}^{(k)} A)$.
8. [HJ85] If $A \in F^{n \times n}$, $B \in F^{m \times m}$, then $\det(A \otimes B) = (\det A)^m (\det B)^n$. (See Section 10.5 for the definition of $A \otimes B$.)
9. [Uhl02] For $A \in F^{n \times n}$, $\det A$ is the unique normalized, alternating, and multilinear function $d: F^{n \times n} \rightarrow F$. That is, $d(I_n) = 1$, $d(A) = -d(A')$, where A' denotes the matrix obtained from A , by interchanging two rows, and d is linear in each row of A , if the remaining rows of A are held fixed.
10. [HJ85] (Cauchy–Binet) Let $A \in F^{n \times k}$, $B \in F^{k \times n}$, and $C = AB$. Then

$$\det C[\alpha, \beta] = \sum_{\gamma} \det A[\alpha, \gamma] \det B[\gamma, \beta],$$

where $\alpha \subseteq \{1, 2, \dots, m\}$, $\beta \subseteq \{1, 2, \dots, n\}$, with $|\alpha| = |\beta| = r$, $1 \leq r \leq \min\{m, k, n\}$, and the sum is taken over all sets $\gamma \subseteq \{1, 2, \dots, k\}$ with $|\gamma| = r$.

11. [HJ85] (Schur Complement). Let $A[\alpha]$ be nonsingular. Then

$$\det A = \det A[\alpha] \det (A[\alpha^c] - A[\alpha^c, \alpha] A[\alpha]^{-1} A[\alpha, \alpha^c]).$$

12. [HJ85] (Jacobi's Theorem) Let A be nonsingular and let $\alpha, \beta \subseteq \{1, 2, \dots, n\}$, with $|\alpha| = |\beta|$. Then

$$\det A^{-1}[\alpha^c, \beta^c] = (-1)^{(\sum_{i \in \alpha} i + \sum_{j \in \beta} j)} \frac{\det A[\beta, \alpha]}{\det A}.$$

In particular, if $\alpha = \beta$. Then $\det A^{-1}[\alpha^c] = \frac{\det A[\alpha]}{\det A}$.

13. [HJ85] (Sylvester's Identity) Let $\alpha \subseteq \{1, 2, \dots, n\}$ with $|\alpha| = k$, and $i, j \in \{1, 2, \dots, n\}$, with $i, j \notin \alpha$. For $A \in F^{n \times n}$, let $B = [b_{ij}] \in F^{(n-k) \times (n-k)}$ be defined by $b_{ij} = \det A[\alpha \cup \{i\}, \alpha \cup \{j\}]$. Then

$$\det B = (\det A[\alpha])^{n-k-1} \det A.$$

Examples:

1. Let $A = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & -3 & 2 \\ 0 & -1 & -2 & -4 \end{bmatrix}$. $S_3(A) = 23$ because $\det A[\{1, 2, 3\}] = \det \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -3 \end{bmatrix} = -3$,

$$\det A[\{1, 2, 4\}] = \det \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & -1 & -4 \end{bmatrix} = -3, \det A[\{1, 3, 4\}] = \det \begin{bmatrix} 1 & 0 & 0 \\ 0 & -3 & 2 \\ 0 & -2 & -4 \end{bmatrix} = 16, \text{ and}$$

$$\det A[\{2, 3, 4\}] = \det \begin{bmatrix} 1 & 0 & 1 \\ 0 & -3 & 2 \\ -1 & -2 & -4 \end{bmatrix} = 13. \text{ From the Laplace expansion on the first column}$$

and $\det A[\{2, 3, 4\}] = 13$, it follows that $S_4(A) = \det A = 13$. Clearly, $S_1(A) = \text{tr } A = -5$.

2. (k th compound) Let $A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 4 \\ 1 & 1 & 0 & 1 \\ 1 & 4 & 0 & 1 \end{bmatrix}$. Then $\det A = 9$, $C_2(A) = \begin{bmatrix} -1 & 0 & 3 & 1 & 4 & 3 \\ 0 & -1 & 0 & -1 & 0 & 1 \\ 3 & -1 & 0 & -4 & -3 & 1 \\ 1 & -1 & -3 & -1 & -4 & 1 \\ 4 & -1 & -3 & -4 & -16 & 1 \\ 3 & 0 & 0 & 0 & -3 & 0 \end{bmatrix}$,

and $\det(C_2(A)) = 729$.

3. (Cauchy-Binet) Let $A = \begin{bmatrix} -1 & 3 & -1 \\ 2 & 0 & 0 \\ i & -4 & 0 \\ 0 & 1+i & 1 \end{bmatrix}$, $B = \begin{bmatrix} 3 & 2 & -3 & 0 \\ 0 & -4 & 4 & 3i \\ 7 & -6i & 5 & 4 \end{bmatrix}$, and $C = AB$.

Then $\det C[\{2, 4\}, \{2, 3\}] = \det A[\{2, 4\}, \{1, 2\}] \det B[\{1, 2\}, \{2, 3\}] + \det A[\{2, 4\}, \{1, 3\}] \det B[\{1, 3\}, \{2, 3\}] + \det A[\{2, 4\}, \{2, 3\}] \det B[\{2, 3\}, \{2, 3\}] = 12 - 44i$.

4. (Schur Complement) Let $A = \begin{bmatrix} a & \mathbf{b}^* \\ \mathbf{b} & C \end{bmatrix}$, where $a \in \mathbb{C}$, $\mathbf{b} \in \mathbb{C}^{n-1}$, and $C \in \mathbb{C}^{(n-1) \times (n-1)}$. If C is nonsingular, then $\det A = (a - \mathbf{b}^* C^{-1} \mathbf{b}) \det C$. If $a \neq 0$, then $\det A = a \det(C - \frac{1}{a} \mathbf{b} \mathbf{b}^*)$.

5. (Jacobi's Theorem) Let $A = \begin{bmatrix} 1 & -2 & 0 \\ 3 & 4 & 0 \\ -1 & 0 & 5 \end{bmatrix}$ and $\alpha = \{2\} = \beta$. By Jacobi's formula, $\det A^{-1}(2) =$

$$\frac{\det A[2]}{\det A} = \frac{4}{50} = \frac{2}{25}. \text{ This can be readily verified by computing } A^{-1} = \begin{bmatrix} \frac{2}{5} & \frac{1}{5} & 0 \\ -\frac{3}{10} & \frac{1}{10} & 0 \\ \frac{2}{25} & \frac{1}{25} & \frac{1}{5} \end{bmatrix}, \text{ and verifying}$$

$$\det A^{-1}[\{1, 3\}] = \frac{2}{25}.$$

6. (Sylvester's Identity) Let $A = \begin{bmatrix} -7 & i & -3 \\ -i & -2 & 1+4i \\ -3 & 1-4i & 5 \end{bmatrix}$ and $\alpha = \{1\}$. Define $B \in \mathbb{C}^{2 \times 2}$, with entries $b_{11} = \det A[\{1, 2\}] = 13$, $b_{12} = \det A[\{1, 2\}, \{1, 3\}] = -7 - 31i$, $b_{21} = \det A[\{1, 3\}, \{1, 2\}] = -7 + 31i$, $b_{22} = \det A[\{1, 3\}] = -44$. Then $-1582 = \det B = (\det A[\{1\}]) \det A = (-7) \det A$, so $\det A = 226$.

4.3 Eigenvalues and Eigenvectors

Definitions:

An element $\lambda \in F$ is an **eigenvalue** of a matrix $A \in F^{n \times n}$ if there exists a nonzero vector $\mathbf{x} \in F^n$ such that $A\mathbf{x} = \lambda\mathbf{x}$. The vector $\mathbf{x}$ is said to be an **eigenvector** of A corresponding to the eigenvalue λ . A nonzero row vector $\mathbf{y}$ is a **left eigenvector** of A , corresponding to the eigenvalue λ , if $\mathbf{y}A = \lambda\mathbf{y}$.

For $A \in F^{n \times n}$, the **characteristic polynomial** of A is given by $p_A(x) = \det(xI - A)$.

The **algebraic multiplicity**, $\alpha(\lambda)$, of $\lambda \in \sigma(A)$ is the number of times the eigenvalue occurs as a root in the characteristic polynomial of A .

The **spectrum** of $A \in F^{n \times n}$, $\sigma(A)$, is the multiset of all eigenvalues of A , with eigenvalue λ appearing $\alpha(\lambda)$ times in $\sigma(A)$.

The **spectral radius** of $A \in \mathbb{C}^{n \times n}$ is $\rho(A) = \max\{|\lambda| : \lambda \in \sigma(A)\}$.

Let $p(x) = c_n x^n + c_{n-1} x^{n-1} + \cdots + c_2 x^2 + c_1 x + c_0$ be a polynomial with coefficients in F . Then $p(A) = c_n A^n + c_{n-1} A^{n-1} + \cdots + c_2 A^2 + c_1 A + c_0 I$.

For $A \in F^{n \times n}$, the **minimal polynomial** of A , $q_A(x)$, is the unique monic polynomial of least degree for which $q_A(A) = 0$.

The vector space $\ker(A - \lambda I)$, for $\lambda \in \sigma(A)$, is the **eigenspace** of $A \in F^{n \times n}$ corresponding to λ , and is denoted by $E_\lambda(A)$.

The **geometric multiplicity**, $\gamma(\lambda)$, of an eigenvalue λ is the dimension of the eigenspace $E_\lambda(A)$.

An eigenvalue λ is **simple** if $\alpha(\lambda) = 1$.

An eigenvalue λ is **semisimple** if $\alpha(\lambda) = \gamma(\lambda)$.

For $K = \mathbb{C}$ or any other algebraically closed field, a matrix $A \in K^{n \times n}$ is **nonderogatory** if $\gamma(\lambda) = 1$ for all $\lambda \in \sigma(A)$, otherwise A is **derogatory**. Over an arbitrary field F , a matrix is nonderogatory (derogatory) if it is nonderogatory (derogatory) over the algebraic closure of F .

For $K = \mathbb{C}$ or any other algebraically closed field, a matrix $A \in K^{n \times n}$ is **nondefective** if every eigenvalue of A is semisimple, otherwise A is **defective**. Over an arbitrary field F , a matrix is nondefective (defective) if it is nondefective (defective) over the algebraic closure of F .

A matrix $A \in F^{n \times n}$ is **diagonalizable** if there exists a nonsingular matrix $B \in F^{n \times n}$, such that $A = BDB^{-1}$ for some diagonal matrix $D \in F^{n \times n}$.

For a monic polynomial $p(x) = x^n + c_{n-1}x^{n-1} + \cdots + c_2x^2 + c_1x + c_0$ with coefficients in F , the $n \times n$ matrix $C(p) = \begin{bmatrix} 0 & 0 & 0 & \cdots & 0 & -c_0 \\ 1 & 0 & 0 & \cdots & 0 & -c_1 \\ 0 & 1 & 0 & \cdots & 0 & -c_2 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 1 & -c_{n-1} \end{bmatrix}$ is called the **companion matrix** of $p(x)$.

Let T be a linear operator on a finite dimensional vector space, V , over a field F . An element $\lambda \in F$ is an **eigenvalue** of T if there exists a nonzero vector $\mathbf{v} \in V$ such that $T(\mathbf{v}) = \lambda\mathbf{v}$. The vector $\mathbf{v}$ is said to be an **eigenvector** of T corresponding to the eigenvalue λ .

For a linear operator, T , on a finite dimensional vector space, V , with a basis, $\mathcal{B}$, the **characteristic polynomial** of T is given by $p_T(x) = \det([T]_{\mathcal{B}})$.

A linear operator T on a finite dimensional vector space, V , is **diagonalizable** if there exists a basis, $\mathcal{B}$, for V such that $[T]_{\mathcal{B}}$ is diagonalizable.

Facts:

These facts are grouped into the following categories: Eigenvalues and Eigenvectors, Diagonalization, Polynomials, Other Facts. All matrices are assumed to be in $F^{n \times n}$ unless otherwise stated. All the following facts, except those with a specific reference, can be found in [Mey00, pp. 489–660] or [Lay03, pp. 301–342].

Eigenvalues and Eigenvectors

1. $\lambda \in \sigma(A)$ if and only if $p_A(\lambda) = 0$.
2. For each eigenvalue λ of a matrix A , $1 \leq \gamma(\lambda) \leq \alpha(\lambda)$.
3. A simple eigenvalue is semisimple.
4. For any F , $|\sigma(A)| \leq n$. If $F = \mathbb{C}$ or any algebraically closed field, then $|\sigma(A)| = n$.
5. If $F = \mathbb{C}$ or any algebraically closed field, then $\det A = \prod_{i=1}^n \lambda_i$, $\lambda_i \in \sigma(A)$.
6. If $F = \mathbb{C}$ or any algebraically closed field, then $\text{tr } A = \sum_{i=1}^n \lambda_i$, $\lambda_i \in \sigma(A)$.
7. For $A \in \mathbb{C}^{n \times n}$, $\lambda \in \sigma(A)$ if and only if $\bar{\lambda} \in \sigma(A^*)$.
8. For $A \in \mathbb{R}^{n \times n}$, viewing $A \in \mathbb{C}^{n \times n}$, $\lambda \in \sigma(A)$ if and only if $\bar{\lambda} \in \sigma(A)$.
9. If $A \in \mathbb{C}^{n \times n}$ is Hermitian (e.g., $A \in \mathbb{R}^{n \times n}$ is symmetric), then A has real eigenvalues and A can be diagonalized. (See also Section 7.2.)
10. A and A^T have the same eigenvalues with same algebraic multiplicities.
11. If $A = [a_{ij}]$ is triangular, then $\sigma(A) = \{a_{11}, a_{22}, \dots, a_{nn}\}$.
12. If A has all row (column) sums equal to r , then r is an eigenvalue of A .
13. A is singular if and only if $\det A = 0$, if and only if $0 \in \sigma(A)$.
14. If A is nonsingular and λ is an eigenvalue of A of algebraic multiplicity $\alpha(\lambda)$, with corresponding eigenvector $\mathbf{x}$, then λ^{-1} is an eigenvalue of A^{-1} with algebraic multiplicity $\alpha(\lambda)$ and corresponding eigenvector $\mathbf{x}$.
15. Let $\lambda_1, \lambda_2, \dots, \lambda_s$ be distinct eigenvalues of A . For each $i = 1, 2, \dots, s$ let $\mathbf{x}_{i1}, \mathbf{x}_{i2}, \dots, \mathbf{x}_{ir_i}$ be linearly independent eigenvectors corresponding to λ_i . Then the vectors $\mathbf{x}_{11}, \dots, \mathbf{x}_{1r_1}, \mathbf{x}_{21}, \dots, \mathbf{x}_{2r_2}, \dots, \mathbf{x}_{s1}, \dots, \mathbf{x}_{sr_s}$ are linearly independent.
16. [FIS89] Let T be a linear operator on a finite dimensional vector space over a field F , with basis $\mathcal{B}$. Then $\lambda \in F$ is an eigenvalue of T if and only if λ is an eigenvalue of $[T]_{\mathcal{B}}$.
17. [FIS89] Let $\lambda_1, \lambda_2, \dots, \lambda_s$ be distinct eigenvalues of the linear operator T , on a finite dimensional space V . For each $i = 1, 2, \dots, s$ let $\mathbf{x}_{i1}, \mathbf{x}_{i2}, \dots, \mathbf{x}_{ir_i}$ be linearly independent eigenvectors corresponding to λ_i . Then the vectors $\mathbf{x}_{11}, \dots, \mathbf{x}_{1r_1}, \mathbf{x}_{21}, \dots, \mathbf{x}_{2r_2}, \dots, \mathbf{x}_{s1}, \dots, \mathbf{x}_{sr_s}$ are linearly independent.
18. Let T be linear operator on a finite dimensional vector space V over a field F . Then $\lambda \in F$ is an eigenvalue of T if and only if $p_T(\lambda) = 0$.

Diagonalization

19. [Lew91, pp. 135–136] Let $\lambda_1, \lambda_2, \dots, \lambda_s$ be distinct eigenvalues of A . If $A \in \mathbb{C}^{n \times n}$, then A is diagonalizable if and only if $\alpha(\lambda_i) = \gamma(\lambda_i)$ for $i = 1, 2, \dots, s$. If $A \in \mathbb{R}^{n \times n}$, then A is diagonalizable by a nonsingular matrix $B \in \mathbb{R}^{n \times n}$ if and only if all the eigenvalues of A are real and $\alpha(\lambda_i) = \gamma(\lambda_i)$ for $i = 1, 2, \dots, s$.
20. Method for Diagonalization of A over $\mathbb{C}$: This is a theoretical method using exact arithmetic and is undesirable in decimal arithmetic with rounding errors. See Chapter 43 for information on appropriate numerical methods.
 - Find the eigenvalues of A .
 - Find a basis $\mathbf{x}_{i1}, \dots, \mathbf{x}_{ir_i}$ for $E_{\lambda_i}(A)$ for each of the distinct eigenvalues $\lambda_1, \dots, \lambda_k$ of A .
 - If $r_1 + \dots + r_k = n$, then let $B = [\mathbf{x}_{11} \dots \mathbf{x}_{1r_1} \dots \mathbf{x}_{k1} \dots \mathbf{x}_{kr_k}]$, B is invertible and $D = B^{-1}AB$ is a diagonal matrix, whose diagonal entries are the eigenvalues of A , in the order that corresponds to the order of the columns of B . Else A is not diagonalizable.
21. A is diagonalizable if and only if A has n linearly independent eigenvectors.

22. A is diagonalizable if and only if $|\sigma(A)| = n$ and A is nondefective.
23. If A has n distinct eigenvalues, then A is diagonalizable.
24. A is diagonalizable if and only if $q_A(x)$ can be factored into distinct linear factors.
25. If A is diagonalizable, then so are $A^T, A^k, k \in \mathbb{N}$.
26. If A is nonsingular and diagonalizable, then A^{-1} is diagonalizable.
27. If A is an idempotent, then A is diagonalizable and $\sigma(A) \subseteq \{0, 1\}$.
28. If A is nilpotent, then $\sigma(A) = \{0\}$. If A is nilpotent and is not the zero matrix, then A is not diagonalizable.
29. [FIS89] Let T be a linear operator on a finite dimensional vector space V with a basis $\mathcal{B}$. Then T is diagonalizable if and only if $[T]_{\mathcal{B}}$ is diagonalizable.
30. [FIS89] A linear operator, T , on a finite dimensional vector space V is diagonalizable if and only if there exists a basis $\mathcal{B} = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ for V , and scalars $\lambda_1, \dots, \lambda_n$, such that $T(\mathbf{v}_i) = \lambda_i \mathbf{v}_i$, for $1 \leq i \leq n$.
31. [FIS89] If a linear operator, T , on a vector space, V , of dimension n , has n distinct eigenvalues, then it is diagonalizable.
32. [FIS89] The characteristic polynomial of a diagonalizable linear operator on a finite dimensional vector space can be factored into linear terms.

Polynomials

33. [HJ85] (Cayley–Hamilton Theorem) Let $p_A(x) = x^n + a_{n-1}x^{n-1} + \dots + a_1x + a_0$ be the characteristic polynomial of A . Then $p_A(A) = A^n + a_{n-1}A^{n-1} + \dots + a_1A + a_0I_n = 0$.
34. [FIS89] (Cayley–Hamilton Theorem for a Linear Operator) Let $p_T(x) = x^n + a_{n-1}x^{n-1} + \dots + a_1x + a_0$ be the characteristic polynomial of a linear operator, T , on a finite dimensional vector space, V . Then $p_T(T) = T^n + a_{n-1}T^{n-1} + \dots + a_1T + a_0I_n = T_0$, where T_0 is the zero linear operator on V .
35. $p_{A^T}(x) = p_A(x)$.
36. The minimal polynomial $q_A(x)$ of a matrix A is a factor of the characteristic polynomial $p_A(x)$ of A .
37. If λ is an eigenvalue of A associated with the eigenvector $\mathbf{x}$, then $p(\lambda)$ is an eigenvalue of the matrix $p(A)$ associated with the eigenvector $\mathbf{x}$, where $p(x)$ is a polynomial with coefficients in F .
38. If B is nonsingular, $p_A(x) = p_{B^{-1}AB}(x)$, therefore, A and $B^{-1}AB$ have the same eigenvalues.
39. Let $p_A(x) = x^n + a_{n-1}x^{n-1} + \dots + a_1x + a_0$ be the characteristic polynomial of A . If $|\sigma(A)| = n$, then $a_k = (-1)^{n-k} S_{n-k}(\lambda_1, \dots, \lambda_n)$, $k = 0, 1, \dots, n-1$, where $S_k(\lambda_1, \dots, \lambda_n)$ is the k th symmetric function of the eigenvalues of A .
40. Let $p_A(x) = x^n + a_{n-1}x^{n-1} + \dots + a_1x + a_0$ be the characteristic polynomial of A . Then $a_k = (-1)^{n-k} S_{n-k}(A)$, $k = 0, 1, \dots, n-1$.
41. If $|\sigma(A)| = n$, then $S_k(A) = S_k(\lambda_1, \dots, \lambda_n)$.
42. If $C(p)$ is the companion matrix of the polynomial $p(x)$, then $p(x) = p_{C(p)}(x) = q_{C(p)}(x)$.
43. [HJ85, p. 135] If $|\sigma(A)| = n$, A is nonderogatory, and B commutes with A , then there exists a polynomial $f(x)$ of degree less than n such that $B = f(A)$.

Other Facts:

44. If A is nonsingular and λ is an eigenvalue of A of algebraic multiplicity $\alpha(\lambda)$, with corresponding eigenvector $\mathbf{x}$, then $\det(A)\lambda^{-1}$ is an eigenvalue of $\text{adj } A$ with algebraic multiplicity $\alpha(\lambda)$ and corresponding eigenvector $\mathbf{x}$.
45. [Lew91] If $\lambda \in \sigma(A)$, then any nonzero column of $\text{adj}(A - \lambda I)$ is an eigenvector of A corresponding to λ .
46. If $AB = BA$, then A and B have a common eigenvector.
47. If $A \in F^{m \times n}$ and $B \in F^{n \times m}$, then $\sigma(AB) = \sigma(BA)$ except for the zero eigenvalues.

48. If $A \in F^{m \times m}$ and $B \in F^{n \times n}$, $\lambda \in \sigma(A)$, $\mu \in \sigma(B)$, with corresponding eigenvectors $\mathbf{u}$ and $\mathbf{v}$, respectively, then $\lambda\mu \in \sigma(A \otimes B)$, with corresponding eigenvector $\mathbf{u} \otimes \mathbf{v}$. (See Section 10.5 for the definition of $A \otimes B$.)

Examples:

1. Let $A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$. Then, viewing $A \in \mathbb{C}^{n \times n}$, $\sigma(A) = \{-i, i\}$. That is, A has no eigenvalues over the reals.

2. Let $A = \begin{bmatrix} -3 & 7 & -1 \\ 6 & 8 & -2 \\ 72 & -28 & 19 \end{bmatrix}$. Then $p_A(x) = (x+6)(x-15)^2 = q_A(x)$, $\lambda_1 = -6$, $\alpha(\lambda_1) = 1$, $\gamma(\lambda_1) = 1$, $\lambda_2 = 15$, $\alpha(\lambda_2) = 2$, $\gamma(\lambda_2) = 1$. Also, a set of linearly independent eigenvectors is $\left\{ \begin{bmatrix} -1 \\ -2 \\ 4 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \\ 4 \end{bmatrix} \right\}$. So, A is not diagonalizable.

3. Let $A = \begin{bmatrix} 57 & -21 & 21 \\ -14 & 22 & -7 \\ -140 & 70 & -55 \end{bmatrix}$. Then $p_A(x) = (x+6)(x-15)^2$, $q_A(x) = (x+6)(x-15)$, $\lambda_1 = -6$, $\alpha(\lambda_1) = 1$, $\gamma(\lambda_1) = 1$, $\lambda_2 = 15$, $\alpha(\lambda_2) = 2$, $\gamma(\lambda_2) = 2$. Also, a set of linearly independent eigenvectors is $\left\{ \begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ 1 \\ 10 \end{bmatrix} \right\}$. So, A is diagonalizable.

4. Let $A = \begin{bmatrix} -5+4i & 1 & i \\ 2+8i & -4 & 2i \\ 20-4i & -4 & -i \end{bmatrix}$. Then $\sigma(A) = \{-6, -3, 3i\}$. If $B = \frac{1}{9}A^2 + 2A - 4I$, then $\sigma(B) = \{-12, -9, -5+6i\}$.

5. Let $A = \begin{bmatrix} -2 & 1 & 0 \\ 0 & -3i & 0 \\ 0 & 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} -2 & 1 & 0 \\ 3 & -1 & 1 \\ 4 & 0 & 1 \end{bmatrix}$. B is nonsingular, so A and $B^{-1}AB = \begin{bmatrix} -1+3i & 1-i & i \\ 5+6i & -1-2i & 1+2i \\ 8-12i & -4+4i & 1-4i \end{bmatrix}$ have the same eigenvalues, which are given in the diagonal of A .

6. Let $A = \begin{bmatrix} 2 & -1 & 0 \\ 0 & 3 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 0 \\ 2 & 1 \\ 1 & 0 \end{bmatrix}$. Then $AB = \begin{bmatrix} 4 & -1 \\ 7 & 3 \end{bmatrix}$, $\sigma(AB) = \left\{ \frac{1}{2}(7+3\sqrt{3}i), \frac{1}{2}(7-3\sqrt{3}i) \right\}$, $BA = \begin{bmatrix} 6 & -3 & 0 \\ 4 & 1 & 1 \\ 2 & -1 & 0 \end{bmatrix}$, and $\sigma(BA) = \left\{ \frac{1}{2}(7+3\sqrt{3}i), \frac{1}{2}(7-3\sqrt{3}i), 0 \right\}$.

7. Let $A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -3i \end{bmatrix}$ and $B = \begin{bmatrix} -2 & 7 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & i \end{bmatrix}$. Then $AB = BA = \begin{bmatrix} -2 & 5 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$. A and B

share the eigenvector $\mathbf{x} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, corresponding to the eigenvalues $\lambda = 1$ of A and $\mu = -2$ of B .

8. Let $A = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & -3 & 2 \\ 0 & -1 & -2 & -4 \end{bmatrix}$. Then $p_A(x) = x^4 + 5x^3 + 4x^2 - 23x + 13$. $S_4(A)$, $S_3(A)$, and $S_1(A)$

were computed in Example 1 of Section 4.2, and it is straightforward to verify that $S_2(A) = 4$. Comparing these values to the characteristic polynomial, $S_4(A) = 13 = (-1)^4 13$, $S_3(A) = 23 = (-1)^3(-23)$, $S_2(A) = (-1)^2 4$, and $S_1(A) = (-1)(5)$. It follows that $S_4(\lambda_1, \lambda_2, \lambda_3, \lambda_4) = \lambda_1 \lambda_2 \lambda_3 \lambda_4 = 13$, $S_3(\lambda_1, \lambda_2, \lambda_3, \lambda_4) = 23$, $S_2(\lambda_1, \lambda_2, \lambda_3, \lambda_4) = 4$, and $S_1(\lambda_1, \lambda_2, \lambda_3, \lambda_4) = \lambda_1 + \lambda_2 + \lambda_3 + \lambda_4 = -5$ (these values can also be verified with a computer algebra system or numerical software).

9. Let $p(x) = x^3 - 7x^2 - 3x + 2$, $C = C(p) = \begin{bmatrix} 0 & 0 & -2 \\ 1 & 0 & 3 \\ 0 & 1 & 7 \end{bmatrix}$. Then $p_C(x) = x^3 - 7x^2 - 3x + 2 =$

$$p(x). \text{ Also, } p_C(C) = -C^3 + 7C^2 + 3C - 2I = - \begin{bmatrix} -2 & -14 & -104 \\ 3 & 19 & 142 \\ 7 & 52 & 383 \end{bmatrix} + 7 \begin{bmatrix} 0 & -2 & -14 \\ 0 & 3 & 19 \\ 1 & 7 & 52 \end{bmatrix} \\ + 3 \begin{bmatrix} 0 & 0 & -2 \\ 1 & 0 & 3 \\ 0 & 1 & 7 \end{bmatrix} - 2 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Applications:

- (Markov Chains) (See also Chapter 54 for more information.) A Markov Chain describes a process in which a system can be in any one of n states: $s_1, s_2, \dots, s_n$. The probability of entering state s_i depends only on the state previously occupied by the system. The **transition probability** of entering state j , given that the system is in state i , is denoted by p_{ij} . The **transition matrix** is the matrix $P = [p_{ij}]$; its rows have sum 1. A (row or column) vector is a **probability vector** if its entries are nonnegative and sum to 1. The probability row vector $\pi^{(k)} = (\pi_1^{(k)}, \pi_2^{(k)}, \dots, \pi_n^{(k)})$, $k \geq 0$, is called the **state vector** of the system at time k if its i th entry is the probability that the system is in state s_i at time k . In particular, when $k = 0$, the state vector is called the **initial state vector** and its i th entry is the probability that the system begins at state s_i . It follows from probability theory that $\pi^{(k+1)} = \pi^{(k)}P$, and thus inductively that $\pi^{(k)} = \pi^{(0)}P^k$. If the entries of some power of P are all positive, then P is said to be

regular. If P is a regular transition matrix, then as $n \rightarrow \infty$, $P^n \rightarrow \begin{bmatrix} \pi_1 & \pi_2 & \cdots & \pi_n \\ \pi_1 & \pi_2 & \cdots & \pi_n \\ \vdots & \vdots & \ddots & \vdots \\ \pi_1 & \pi_2 & \cdots & \pi_n \end{bmatrix}$. The

row vector $\pi = (\pi_1, \pi_2, \dots, \pi_n)$ is called the **steady state vector**, π is a probability vector, and as $n \rightarrow \infty$, $\pi^{(n)} \rightarrow \pi$. The vector π is the unique probability row vector with the property that $\pi P = \pi$. That is, π is the unique probability row vector that is a left eigenvector of P for eigenvalue 1.

- (Differential Equations) [Mey00, pp. 541–546] Consider the system of linear differential equations

$$\begin{cases} x_1' = a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n \\ x_2' = a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n \\ \vdots \\ x_n' = a_{n1}x_1 + a_{n2}x_2 + \dots + a_{nn}x_n \end{cases}, \text{ where each of the unknowns } x_1, x_2, \dots, x_n$$

is a differentiable function of the real variable t . This system of linear differential equations can

be written in matrix form as $\mathbf{x}' = A\mathbf{x}$, where $A = [a_{ij}]$, $\mathbf{x} = \begin{bmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_n(t) \end{bmatrix}$, and $\mathbf{x}' = \begin{bmatrix} x'_1(t) \\ x'_2(t) \\ \vdots \\ x'_n(t) \end{bmatrix}$. If A

is diagonalizable, there exists a nonsingular matrix B (the columns of the matrix B are linearly independent eigenvectors of A), such that $B^{-1}AB = D$ is a diagonal matrix, so $\mathbf{x}' = BDB^{-1}\mathbf{x}$, or $B^{-1}\mathbf{x}' = DB^{-1}\mathbf{x}$. Let $\mathbf{u} = B^{-1}\mathbf{x}$. The linear system of differential equations $\mathbf{u}' = D\mathbf{u}$ has solution

$\mathbf{u} = \begin{bmatrix} k_1 e^{\lambda_1 t} \\ k_2 e^{\lambda_2 t} \\ \vdots \\ k_n e^{\lambda_n t} \end{bmatrix}$, where $\lambda_1, \lambda_2, \dots, \lambda_n$ are the eigenvalues of A . It follows that $\mathbf{x} = B\mathbf{u}$. (See also

Chapter 55.)

3. (Dynamical Systems) [Lay03, pp. 315–316] Consider the dynamical system given by $\mathbf{u}_{k+1} = A\mathbf{u}_k$,

where $A = [a_{ij}]$, $\mathbf{u}_0 = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}$. If A is diagonalizable, there exist n linearly independent eigenvectors,

$\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n$, of A . The vector $\mathbf{u}_0$ can then be written as a linear combination of the eigenvectors, that is, $\mathbf{u}_0 = c_1\mathbf{x}_1 + c_2\mathbf{x}_2 + \dots + c_n\mathbf{x}_n$. Then $\mathbf{u}_1 = A\mathbf{u}_0 = A(c_1\mathbf{x}_1 + c_2\mathbf{x}_2 + \dots + c_n\mathbf{x}_n) = c_1\lambda_1\mathbf{x}_1 + c_2\lambda_2\mathbf{x}_2 + \dots + c_n\lambda_n\mathbf{x}_n$. Inductively, $\mathbf{u}_{k+1} = A\mathbf{u}_k = c_1\lambda_1^{k+1}\mathbf{x}_1 + c_2\lambda_2^{k+1}\mathbf{x}_2 + \dots + c_n\lambda_n^{k+1}\mathbf{x}_n$. Thus, the long-term behavior of the dynamical system can be studied using the eigenvalues of the matrix A . (See also Chapter 56.)

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5

Inner Product Spaces, Orthogonal Projection, Least Squares, and Singular Value Decomposition

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5.1 Inner Product Spaces

Definitions:

Let V be a vector space over the field F , where $F = \mathbb{R}$ or $F = \mathbb{C}$. An **inner product** on V is a function $\langle \cdot, \cdot \rangle: V \times V \rightarrow F$ such that for all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$ and $a, b \in F$, the following hold:

- $\langle \mathbf{v}, \mathbf{v} \rangle \geq 0$ and $\langle \mathbf{v}, \mathbf{v} \rangle = 0$ if and only if $\mathbf{v} = \mathbf{0}$.
- $\langle a\mathbf{u} + b\mathbf{v}, \mathbf{w} \rangle = a\langle \mathbf{u}, \mathbf{w} \rangle + b\langle \mathbf{v}, \mathbf{w} \rangle$.
- For $F = \mathbb{R}$: $\langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle$; For $F = \mathbb{C}$: $\langle \mathbf{u}, \mathbf{v} \rangle = \overline{\langle \mathbf{v}, \mathbf{u} \rangle}$ (where bar denotes complex conjugation).

A real (or complex) **inner product space** is a vector space V over $\mathbb{R}$ (or $\mathbb{C}$), together with an inner product defined on it.

In an inner product space V , the **norm**, or **length**, of a vector $\mathbf{v} \in V$ is $\|\mathbf{v}\| = \sqrt{\langle \mathbf{v}, \mathbf{v} \rangle}$.

A vector $\mathbf{v} \in V$ is a **unit vector** if $\|\mathbf{v}\| = 1$.

The **angle** between two nonzero vectors $\mathbf{u}$ and $\mathbf{v}$ in a real inner product space is the real number θ , $0 \leq \theta \leq \pi$, such that $\langle \mathbf{u}, \mathbf{v} \rangle = \|\mathbf{u}\| \|\mathbf{v}\| \cos \theta$. See the Cauchy–Schwarz inequality (Fact 9 below).

Let V be an inner product space. The **distance** between two vectors $\mathbf{u}$ and $\mathbf{v}$ is $d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\|$.

A Hermitian matrix A is **positive definite** if $\mathbf{x}^* A \mathbf{x} > 0$ for all nonzero $\mathbf{x} \in \mathbb{C}^n$. (See Chapter 8 for more information on positive definite matrices.)

Facts:

All the following facts except those with a specific reference can be found in [Rom92, pp. 157–164].

1. The vector space $\mathbb{R}^n$ is an inner product space under the **standard inner product**, or **dot product**, defined by

$$\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u}^T \mathbf{v} = \sum_{i=1}^n u_i v_i.$$

This inner product space is often called n -dimensional **Euclidean space**.

2. The vector space $\mathbb{C}^n$ is an inner product space under the **standard inner product**, defined by

$$\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{v}^* \mathbf{u} = \sum_{i=1}^n u_i \bar{v}_i.$$

This inner product space is often called n -dimensional **unitary space**.

3. [HJ85, p. 410] In $\mathbb{R}^n$, a function $\langle \cdot, \cdot \rangle: \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$ is an inner product if and only if there exists a real symmetric positive definite matrix G such that $\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u}^T G \mathbf{v}$, for all $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$.
4. [HJ85, p. 410] In $\mathbb{C}^n$, a function $\langle \cdot, \cdot \rangle: \mathbb{C}^n \times \mathbb{C}^n \rightarrow \mathbb{C}$ is an inner product if and only if there exists a Hermitian positive definite matrix H such that $\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{v}^* H \mathbf{u}$, for all $\mathbf{u}, \mathbf{v} \in \mathbb{C}^n$.
5. Let l^2 be the vector space of all infinite complex sequences $\mathbf{v} = (v_n)$ with the property that $\sum_{n=1}^{\infty} |v_n|^2 < \infty$. Then l^2 is an inner product space under the inner product

$$\langle \mathbf{u}, \mathbf{v} \rangle = \sum_{n=1}^{\infty} u_n \bar{v}_n.$$

6. The vector space $C[a, b]$ of all continuous real-valued functions on the closed interval $[a, b]$ is an inner product space under the inner product

$$\langle f, g \rangle = \int_a^b f(x)g(x)dx.$$

7. If V is an inner product space and $\langle \mathbf{u}, \mathbf{w} \rangle = \langle \mathbf{v}, \mathbf{w} \rangle$ for all $\mathbf{w} \in V$, then $\mathbf{u} = \mathbf{v}$.
8. The inner product on an inner product space V , when restricted to vectors in a subspace S of V , is an inner product on S .
9. Let V be an inner product space. Then the norm function $\|\cdot\|$ on V has the following basic properties for all $\mathbf{u}, \mathbf{v} \in V$:
 - $\|\mathbf{v}\| \geq 0$ and $\|\mathbf{v}\| = 0$ if and only if $\mathbf{v} = \mathbf{0}$.
 - $\|a\mathbf{v}\| = |a|\|\mathbf{v}\|$, for all $a \in F$.
 - (The triangle inequality) $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$ with equality if and only if $\mathbf{v} = a\mathbf{u}$, for some $a \in F$.
 - (The Cauchy–Schwarz inequality) $|\langle \mathbf{u}, \mathbf{v} \rangle| \leq \|\mathbf{u}\|\|\mathbf{v}\|$ with equality if and only if $\mathbf{v} = a\mathbf{u}$, for some $a \in F$.
 - $|\|\mathbf{u}\| - \|\mathbf{v}\|| \leq \|\mathbf{u} - \mathbf{v}\|$.

- (The parallelogram law) $\|\mathbf{u} + \mathbf{v}\|^2 + \|\mathbf{u} - \mathbf{v}\|^2 = 2\|\mathbf{u}\|^2 + 2\|\mathbf{v}\|^2$.
- (Polarization identities)

$$4\langle \mathbf{u}, \mathbf{v} \rangle = \begin{cases} \|\mathbf{u} + \mathbf{v}\|^2 - \|\mathbf{u} - \mathbf{v}\|^2, & \text{if } F = \mathbb{R}. \\ \|\mathbf{u} + \mathbf{v}\|^2 - \|\mathbf{u} - \mathbf{v}\|^2 + i\|\mathbf{u} + i\mathbf{v}\|^2 - i\|\mathbf{u} - i\mathbf{v}\|^2, & \text{if } F = \mathbb{C}. \end{cases}$$

Examples:

1. Let $\mathbb{R}^4$ be the Euclidean space with the inner product $\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u}^T \mathbf{v}$. Let $\mathbf{x} = [1, 2, 3, 4]^T \in \mathbb{R}^4$ and $\mathbf{y} = [3, -1, 0, 2]^T \in \mathbb{R}^4$ be two vectors. Then
 - $\langle \mathbf{x}, \mathbf{y} \rangle = 9$, $\|\mathbf{x}\| = \sqrt{30}$, and $\|\mathbf{y}\| = \sqrt{14}$.
 - The distance between $\mathbf{x}$ and $\mathbf{y}$ is $d(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\| = \sqrt{26}$.
 - The angle between $\mathbf{x}$ and $\mathbf{y}$ is $\theta = \arccos \frac{9}{\sqrt{30}\sqrt{14}} = \arccos \frac{9}{2\sqrt{105}} \approx 1.116$ radians.
2. $\langle \mathbf{u}, \mathbf{v} \rangle = u_1v_1 + 2u_1v_2 + 2u_2v_1 + 6u_2v_2 = \mathbf{u}^T \begin{bmatrix} 1 & 2 \\ 2 & 6 \end{bmatrix} \mathbf{v}$ is an inner product on $\mathbb{R}^2$, as the matrix $G = \begin{bmatrix} 1 & 2 \\ 2 & 6 \end{bmatrix}$ is symmetric positive definite.
3. Let $C[-1, 1]$ be the vector space with the inner product $\langle f, g \rangle = \int_{-1}^1 f(x)g(x)dx$ and let $f(x) = 1$ and $g(x) = x^2$ be two functions in $C[-1, 1]$. Then $\langle f, g \rangle = \int_{-1}^1 x^2 dx = 2/3$, $\langle f, f \rangle = \int_{-1}^1 1 dx = 2$, and $\langle g, g \rangle = \int_{-1}^1 x^4 dx = 2/5$. The angle between f and g is $\arccos(\sqrt{5}/3) \approx 0.730$ radians.
4. [Mey00, p. 286] $\langle A, B \rangle = \text{tr}(AB^*)$ is an inner product on $\mathbb{C}^{m \times n}$.

5.2 Orthogonality

Definitions:

Let V be an inner product space. Two vectors $\mathbf{u}, \mathbf{v} \in V$ are **orthogonal** if $\langle \mathbf{u}, \mathbf{v} \rangle = 0$, and this is denoted by $\mathbf{u} \perp \mathbf{v}$.

A subset S of an inner product space V is an **orthogonal set** if $\mathbf{u} \perp \mathbf{v}$, for all $\mathbf{u}, \mathbf{v} \in S$ such that $\mathbf{u} \neq \mathbf{v}$.

A subset S of an inner product space V is an **orthonormal set** if S is an orthogonal set and each $\mathbf{v} \in S$ is a unit vector.

Two subsets S and W of an inner product space V are **orthogonal** if $\mathbf{u} \perp \mathbf{v}$, for all $\mathbf{u} \in S$ and $\mathbf{v} \in W$, and this is denoted by $S \perp W$.

The **orthogonal complement** of a subset S of an inner product space V is $S^\perp = \{\mathbf{w} \in V \mid \langle \mathbf{w}, \mathbf{v} \rangle = 0 \text{ for all } \mathbf{v} \in S\}$.

A **complete orthonormal set** M in an inner product space V is an orthonormal set of vectors in V such that for $\mathbf{v} \in V$, $\mathbf{v} \perp M$ implies that $\mathbf{v} = \mathbf{0}$.

An **orthogonal basis** for an inner product space V is an orthogonal set that is also a basis for V .

An **orthonormal basis** for V is an orthonormal set that is also a basis for V .

A matrix U is **unitary** if $U^*U = I$.

A real matrix Q is **orthogonal** if $Q^T Q = I$.

Facts:

1. [Mey00, p. 298] An orthogonal set of nonzero vectors is linearly independent. An orthonormal set of vectors is linearly independent.
2. [Rom92, p. 164] If S is a subset of an inner product space V , then $S^\perp$ is a subspace of V . Moreover, if S is a subspace of V , then $S \cap S^\perp = \{\mathbf{0}\}$.
3. [Mey00, p. 409] In an inner product space V , $\{\mathbf{0}\}^\perp = V$ and $V^\perp = \{\mathbf{0}\}$.
4. [Rom92, p. 168] If S is a finite dimensional subspace of an inner product space V , then for any $\mathbf{v} \in V$,
 - There are unique vectors $\mathbf{s} \in S$ and $\mathbf{t} \in S^\perp$ such that $\mathbf{v} = \mathbf{s} + \mathbf{t}$. This implies $V = S \oplus S^\perp$.
 - There is a unique linear operator P such that $P(\mathbf{v}) = \mathbf{s}$.
5. [Mey00, p. 404] If S is a subspace of an n -dimensional inner product space V , then
 - $(S^\perp)^\perp = S$.
 - $\dim(S^\perp) = n - \dim(S)$.
6. [Rom92, p. 174] If S is a subspace of an infinite dimensional inner product space, then $S \subseteq (S^\perp)^\perp$, but the two sets need not be equal.
7. [Rom92, p. 166] An orthonormal basis is a complete orthonormal set.
8. [Rom92, p. 166] In a finite-dimensional inner product space, a complete orthonormal set is a basis.
9. [Rom92, p. 165] In an infinite-dimensional inner product space, a complete orthonormal set may not be a basis.
10. [Rom92, p. 166] Every finite-dimensional inner product space has an orthonormal basis.
11. [Mey00, p. 299] Let $\mathcal{B} = \{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n\}$ be an orthonormal basis for V . Every vector $\mathbf{v} \in V$ can be uniquely expressed as

$$\mathbf{v} = \sum_{i=1}^n \langle \mathbf{v}, \mathbf{u}_i \rangle \mathbf{u}_i.$$

The expression on the right is called the **Fourier expansion** of $\mathbf{v}$ with respect to $\mathcal{B}$ and the scalars $\langle \mathbf{v}, \mathbf{u}_i \rangle$ are called the **Fourier coefficients**.

12. [Mey00, p. 305] (Pythagorean Theorem) If $\{\mathbf{v}_i\}_{i=1}^k$ is an orthogonal set of vectors in V , then $\|\sum_{i=1}^k \mathbf{v}_i\|^2 = \sum_{i=1}^k \|\mathbf{v}_i\|^2$.
13. [Rom92, p. 167] (Bessel's Inequality) If $\{\mathbf{u}_i\}_{i=1}^k$ is an orthonormal set of vectors in V , then $\|\mathbf{v}\|^2 \geq \sum_{i=1}^k |\langle \mathbf{v}, \mathbf{u}_i \rangle|^2$.
14. [Mey00, p. 305] (Parseval's Identity) Let $\mathcal{B} = \{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n\}$ be an orthonormal basis for V . Then for each $\mathbf{v} \in V$, $\|\mathbf{v}\|^2 = \sum_{i=1}^n |\langle \mathbf{v}, \mathbf{u}_i \rangle|^2$.
15. [Mey00, p. 405] Let $A \in F^{m \times n}$, where $F = \mathbb{R}$ or $\mathbb{C}$. Then
 - $\ker(A)^\perp = \text{range}(A^*)$, $\text{range}(A)^\perp = \ker(A^*)$.
 - $F^m = \text{range}(A) \oplus \text{range}(A)^\perp = \text{range}(A) \oplus \ker(A^*)$.
 - $F^n = \ker(A) \oplus \ker(A)^\perp = \ker(A) \oplus \text{range}(A^*)$.
16. [Mey00, p. 321] (See also Section 7.1.) The following statements for a real matrix $Q \in \mathbb{R}^{n \times n}$ are equivalent:
 - Q is orthogonal.
 - Q has orthonormal columns.
 - Q has orthonormal rows.
 - $QQ^T = I$, where I is the identity matrix of order n .
 - For all $\mathbf{v} \in \mathbb{R}^n$, $\|Q\mathbf{v}\| = \|\mathbf{v}\|$.

17. [Mey00, p. 321] (See also Section 7.1.) The following statements for a complex matrix $U \in \mathbb{C}^{n \times n}$ are equivalent:

- U is unitary.
- U has orthonormal columns.
- U has orthonormal rows.
- $UU^* = I$, where I is the identity matrix of order n .
- For all $\mathbf{v} \in \mathbb{C}^n$, $\|U\mathbf{v}\| = \|\mathbf{v}\|$.

Examples:

1. Let $C[-1, 1]$ be the vector space with the inner product $\langle f, g \rangle = \int_{-1}^1 f(x)g(x)dx$ and let $f(x) = 1$ and $g(x) = x$ be two functions in $C[-1, 1]$. Then $\langle f, g \rangle = \int_{-1}^1 x dx = 0$. Thus, $f \perp g$.
2. The standard basis $\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n\}$ is an orthonormal basis for the unitary space $\mathbb{C}^n$.
3. If $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ is an orthogonal basis for $\mathbb{C}^n$ and $S = \text{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ ($1 \leq k \leq n-1$), then $S^\perp = \text{span}\{\mathbf{v}_{k+1}, \dots, \mathbf{v}_n\}$.
4. The vectors $\mathbf{v}_1 = [2, 2, 1]^T$, $\mathbf{v}_2 = [1, -1, 0]^T$, and $\mathbf{v}_3 = [-1, -1, 4]^T$ are mutually orthogonal. They can be normalized to $\mathbf{u}_1 = \mathbf{v}_1/\|\mathbf{v}_1\| = [2/3, 2/3, 1/3]^T$, $\mathbf{u}_2 = \mathbf{v}_2/\|\mathbf{v}_2\| = [1/\sqrt{2}, -1/\sqrt{2}, 0]^T$, and $\mathbf{u}_3 = \mathbf{v}_3/\|\mathbf{v}_3\| = [-\sqrt{2}/6, -\sqrt{2}/6, 2\sqrt{2}/3]^T$. The set $\mathbb{B} = \{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ forms an orthonormal basis for the Euclidean space $\mathbb{R}^3$.
 - If $\mathbf{v} = [v_1, v_2, v_3]^T \in \mathbb{R}^3$, then $\mathbf{v} = \langle \mathbf{v}, \mathbf{u}_1 \rangle \mathbf{u}_1 + \langle \mathbf{v}, \mathbf{u}_2 \rangle \mathbf{u}_2 + \langle \mathbf{v}, \mathbf{u}_3 \rangle \mathbf{u}_3$, that is,

$$\mathbf{v} = \frac{2v_1 + 2v_2 + v_3}{3} \mathbf{u}_1 + \frac{v_1 - v_2}{\sqrt{2}} \mathbf{u}_2 + \frac{-v_1 - v_2 + 4v_3}{3\sqrt{2}} \mathbf{u}_3.$$

- The matrix $Q = [\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3] \in \mathbb{R}^{3 \times 3}$ is an orthogonal matrix.
5. Let S be the subspace of $\mathbb{C}^3$ spanned by the vectors $\mathbf{u} = [i, 1, 1]^T$ and $\mathbf{v} = [1, i, 1]^T$. Then the orthogonal complement of S is

$$S^\perp = \{\mathbf{w} | \mathbf{w} = \alpha[1, 1, -1+i]^T, \text{ where } \alpha \in \mathbb{C}\}.$$

6. Consider the inner product space l^2 from Fact 5 in Section 5.1. Let $\mathcal{E} = \{\mathbf{e}_i | i = 1, 2, \dots\}$, where $\mathbf{e}_i$ has a 1 on i th place and 0s elsewhere. It is clear that $\mathcal{E}$ is an orthonormal set. If $\mathbf{v} = (v_n) \perp \mathcal{E}$, then for each n , $v_n = \langle \mathbf{v}, \mathbf{e}_n \rangle = 0$. This implies $\mathbf{v} = \mathbf{0}$. Therefore, $\mathcal{E}$ is a complete orthonormal set. However, $\mathcal{E}$ is not a basis for l^2 as $S = \text{span}\{\mathcal{E}\} \neq l^2$. Further, $S^\perp = \{\mathbf{0}\}$. Thus, $(S^\perp)^\perp = l^2 \not\subseteq S$ and $l^2 \neq S \oplus S^\perp$.

5.3 Adjoint of Linear Operators on Inner Product Spaces

Let V be a finite dimensional (real or complex) inner product space and let T be a linear operator on V .

Definitions:

A linear operator T^* on V is called the **adjoint** of T if $\langle T(\mathbf{u}), \mathbf{v} \rangle = \langle \mathbf{u}, T^*(\mathbf{v}) \rangle$ for all $\mathbf{u}, \mathbf{v} \in V$.

The linear operator T is **self-adjoint**, or **Hermitian**, if $T = T^*$; T is **unitary** if $T^*T = I_V$.

Facts:

The following facts can be found in [HK71].

1. Let f be a linear functional on V . Then there exists a unique $\mathbf{v} \in V$ such that $f(\mathbf{w}) = \langle \mathbf{w}, \mathbf{v} \rangle$ for all $\mathbf{w} \in V$.
2. The adjoint T^* of T exists and is unique.
3. Let $\mathcal{B} = (\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n)$ be an ordered, orthonormal basis of V . Let $A = [T]_{\mathcal{B}}$. Then

$$a_{ij} = \langle T(\mathbf{u}_j), \mathbf{u}_i \rangle, \quad i, j = 1, 2, \dots, n.$$

Moreover, $[T^*]_{\mathcal{B}} = A^*$, the Hermitian adjoint of A .

4. (Properties of the adjoint operator)
 - (a) $(T^*)^* = T$ for every linear operator T on V .
 - (b) $(aT)^* = \bar{a}T^*$ for every linear operator T on V and every $a \in F$.
 - (c) $(T + T_1)^* = T^* + T_1^*$ for every linear operators T, T_1 on V .
 - (d) $(TT_1)^* = T_1^*T^*$ for every linear operators T, T_1 on V .
5. Let $\mathcal{B}$ be an ordered orthonormal basis of V and let $A = [T]_{\mathcal{B}}$. Then
 - (a) T is self-adjoint if and only if A is a Hermitian matrix.
 - (b) T is unitary if and only if A is a unitary matrix.

Examples:

1. Consider the space $\mathbb{R}^3$ equipped with the standard inner product and let $f(\mathbf{w}) = 3w_1 - 2w_3$. Then with $\mathbf{v} = [3, 0, -2]^T$, $f(\mathbf{w}) = \langle \mathbf{w}, \mathbf{v} \rangle$.
2. Consider the space $\mathbb{R}^3$ equipped with the standard inner product. Let $\mathbf{v} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ and $T(\mathbf{v}) = \begin{bmatrix} 2x + y \\ y - 3z \\ x + y + z \end{bmatrix}$. Then $[T] = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 1 & -3 \\ 1 & 1 & 1 \end{bmatrix}$, so $[T]^* = \begin{bmatrix} 2 & 0 & 1 \\ 1 & 1 & 1 \\ 0 & -3 & 1 \end{bmatrix}$, and $T^*(\mathbf{v}) = \begin{bmatrix} 2x + z \\ x + y + z \\ -3y + z \end{bmatrix}$.
3. Consider the space $\mathbb{C}^{n \times n}$ equipped with the inner product in Example 4 of section 5.1. Let $A, B \in \mathbb{C}^{n \times n}$ and let T be the linear operator on $\mathbb{C}^{n \times n}$ defined by $T(X) = AX + XB$, $X \in \mathbb{C}^{n \times n}$. Then $T^*(X) = A^*X + XB^*$, $X \in \mathbb{C}^{n \times n}$.
4. Let V be an inner product space and let T be a linear operator on V . For a fixed $\mathbf{u} \in V$, $f(\mathbf{w}) = \langle T(\mathbf{w}), \mathbf{u} \rangle$ is a linear functional. By Fact 1, there is a unique vector $\mathbf{v}$ such that $f(\mathbf{w}) = \langle \mathbf{w}, \mathbf{v} \rangle$. Then $T^*(\mathbf{u}) = \mathbf{v}$.

5.4 Orthogonal Projection

Definitions:

Let S be a finite-dimensional subspace of an inner product space V . Then according to Fact 4 in Section 5.2, each $\mathbf{v} \in V$ can be written uniquely as $\mathbf{v} = \mathbf{s} + \mathbf{t}$, where $\mathbf{s} \in S$ and $\mathbf{t} \in S^\perp$. The vector $\mathbf{s}$ is called the **orthogonal projection** of $\mathbf{v}$ onto S and is often written as $\text{Proj}_S \mathbf{v}$, where the linear operator Proj_S is called the **orthogonal projection** onto S along $S^\perp$. When $V = \mathbb{C}^n$ or $V = \mathbb{R}^n$ with the standard inner product, the linear operator Proj_S is often identified with its standard matrix $[\text{Proj}_S]$ and Proj_S is used to denote both the operator and the matrix.

Facts:

1. An orthogonal projection is a projection (as defined in Section 3.6).
2. [Mey00, p. 433] Suppose that P is a projection. The following statements are equivalent:
 - P is an orthogonal projection.
 - $P^* = P$.
 - $\text{range}(P) \perp \ker(P)$.
3. [Mey00, p. 430] If S is a subspace of a finite dimensional inner product space V , then

$$\text{Proj}_{S^\perp} = I - \text{Proj}_S.$$

4. [Mey00, p. 430] Let S be a p -dimensional subspace of the standard inner product space $\mathbb{C}^n$, and let the columns of matrices $M \in \mathbb{C}^{n \times p}$ and $N \in \mathbb{C}^{n \times (n-p)}$ be bases for S and $S^\perp$, respectively. Then the orthogonal projections onto S and $S^\perp$ are

$$\text{Proj}_S = M(M^*M)^{-1}M^* \quad \text{and} \quad \text{Proj}_{S^\perp} = N(N^*N)^{-1}N^*.$$

If M and N contain orthonormal bases for S and $S^\perp$, then $\text{Proj}_S = MM^*$ and $\text{Proj}_{S^\perp} = NN^*$.

5. [Lay03, p. 399] If $\{\mathbf{u}_1, \dots, \mathbf{u}_p\}$ is an orthonormal basis for a subspace S of $\mathbb{C}^n$, then for any $\mathbf{v} \in \mathbb{C}^n$,

$$\text{Proj}_S \mathbf{v} = (\mathbf{u}_1^* \mathbf{v}) \mathbf{u}_1 + \dots + (\mathbf{u}_p^* \mathbf{v}) \mathbf{u}_p.$$

6. [TB97, p. 46] Let $\mathbf{v} \in \mathbb{C}^n$ be a nonzero vector. Then
 - $\text{Proj}_{\mathbf{v}} = \frac{\mathbf{v}\mathbf{v}^*}{\mathbf{v}^*\mathbf{v}}$ is the orthogonal projection onto the line $L = \text{span}\{\mathbf{v}\}$.
 - $\text{Proj}_{\perp \mathbf{v}} = I - \frac{\mathbf{v}\mathbf{v}^*}{\mathbf{v}^*\mathbf{v}}$ is the orthogonal projection onto $L^\perp$.
7. [Mey00, p. 435] (The Best Approximation Theorem) Let S be a finite dimensional subspace of an inner product space V and let $\mathbf{b}$ be a vector in V . Then $\text{Proj}_S \mathbf{b}$ is the unique vector in S that is closest to $\mathbf{b}$ in the sense that

$$\min_{\mathbf{s} \in S} \|\mathbf{b} - \mathbf{s}\| = \|\mathbf{b} - \text{Proj}_S \mathbf{b}\|.$$

The vector $\text{Proj}_S \mathbf{b}$ is called **the best approximation to $\mathbf{b}$ by the elements of S** .

Examples:

1. Generally, an orthogonal projection $P \in \mathbb{C}^{n \times n}$ is not a unitary matrix.
2. Let $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ be an orthogonal basis for $\mathbb{R}^n$ and let S the subspace of $\mathbb{R}^n$ spanned by $\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$, where $1 \leq k \leq n-1$. Then $\mathbf{w} = c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \dots + c_n \mathbf{v}_n \in \mathbb{R}^n$ can be written as $\mathbf{w} = \mathbf{s} + \mathbf{t}$, where $\mathbf{s} = c_1 \mathbf{v}_1 + \dots + c_k \mathbf{v}_k \in S$ and $\mathbf{t} = c_{k+1} \mathbf{v}_{k+1} + \dots + c_n \mathbf{v}_n \in S^\perp$.
3. Let $\mathbf{u}_1 = [2/3, 2/3, 1/3]^T$, $\mathbf{u}_2 = [1/3, -2/3, 2/3]^T$, and $\mathbf{x} = [4, 2, 3]^T$. Then $\{\mathbf{u}_1, \mathbf{u}_2\}$ is an orthonormal basis for the subspace $S = \text{span}\{\mathbf{u}_1, \mathbf{u}_2\}$ of $\mathbb{R}^3$.
 - The orthogonal projection of $\mathbf{x}$ onto S is

$$\text{Proj}_S \mathbf{x} = \left(\mathbf{u}_1^T \right) \mathbf{x} \mathbf{u}_1 + \left(\mathbf{u}_2^T \right) \mathbf{x} \mathbf{u}_2 = [4, 2, 3]^T.$$

- The orthogonal projection of $\mathbf{x}$ onto $S^\perp$ is $\mathbf{y} = \mathbf{x} - \text{Proj}_S \mathbf{x} = [-2, 1, 2]^T$.
- The vector in S that is closest to $\mathbf{x}$ is $\text{Proj}_S \mathbf{x} = [4, 2, 3]^T$.

- Let $M = [\mathbf{u}_1, \mathbf{u}_2]$. Then the orthogonal projection onto S is

$$\text{Proj}_S = MM^T = \frac{1}{9} \begin{bmatrix} 5 & 2 & 4 \\ 2 & 8 & -2 \\ 4 & -2 & 5 \end{bmatrix}.$$

- The orthogonal projection of any $\mathbf{v} \in \mathbb{R}^3$ onto S can be computed by $\text{Proj}_S \mathbf{v} = MM^T \mathbf{v}$. In particular, $MM^T \mathbf{x} = [4, 2, 3]^T$.
4. Let $\mathbf{w}_1 = [1, 1, 0]^T$ and $\mathbf{w}_2 = [1, 0, 1]^T$. Consider the subspace $W = \text{span}\{\mathbf{w}_1, \mathbf{w}_2\}$ of $\mathbb{R}^3$. Define

the matrix $M = [\mathbf{w}_1, \mathbf{w}_2] = \begin{bmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$. Then $M^T M = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$.

- The orthogonal projection onto W is $\text{Proj}_W = M(M^T M)^{-1} M^T =$

$$\begin{bmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}^{-1} \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}.$$

- The orthogonal projection of any $\mathbf{v} \in \mathbb{R}^3$ onto W can be computed by $\text{Proj}_W \mathbf{v}$. For $\mathbf{v} = [1, 2, 3]^T$, $\text{Proj}_W \mathbf{v} = \text{Proj}_W [1, 2, 3]^T = [7/3, 2/3, 5/3]^T$.

5.5 Gram–Schmidt Orthogonalization and QR Factorization

Definitions:

Let $\{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n\}$ be a basis for a subspace S of an inner product space V . An orthonormal basis $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n\}$ for S can be constructed using the following Gram–Schmidt orthogonalization process:

$$\mathbf{u}_1 = \frac{\mathbf{a}_1}{\|\mathbf{a}_1\|} \quad \text{and} \quad \mathbf{u}_k = \frac{\mathbf{a}_k - \sum_{i=1}^{k-1} \langle \mathbf{a}_k, \mathbf{u}_i \rangle \mathbf{u}_i}{\|\mathbf{a}_k - \sum_{i=1}^{k-1} \langle \mathbf{a}_k, \mathbf{u}_i \rangle \mathbf{u}_i\|}, \quad \text{for } k = 2, \dots, n.$$

A **reduced QR factorization** of $A \in \mathbb{C}^{m \times n}$ ($m \geq n$) is a factorization $A = \hat{Q} \hat{R}$, where $\hat{Q} \in \mathbb{C}^{m \times n}$ has orthonormal columns and $\hat{R} \in \mathbb{C}^{n \times n}$ is an upper triangular matrix.

A **QR factorization** of $A \in \mathbb{C}^{m \times n}$ ($m \geq n$) is a factorization $A = QR$, where $Q \in \mathbb{C}^{m \times m}$ is a unitary matrix and $R \in \mathbb{C}^{m \times n}$ is an upper triangular matrix with the last $m - n$ rows of R being zero.

Facts:

1. [TB97, p. 51] Each $A \in \mathbb{C}^{m \times n}$ ($m \geq n$) has a full QR factorization $A = QR$. If $A \in \mathbb{R}^{m \times n}$, then both Q and R may be taken to be real.
2. [TB97, p. 52] Each $A \in \mathbb{C}^{m \times n}$ ($m \geq n$) has a reduced QR factorization $A = \hat{Q} \hat{R}$. If $A \in \mathbb{R}^{m \times n}$, then both $\hat{Q}$ and $\hat{R}$ may be taken to be real.
3. [TB97, p. 52] Each $A \in \mathbb{C}^{m \times n}$ ($m \geq n$) of full rank has a unique reduced QR factorization $A = \hat{Q} \hat{R}$, where $\hat{Q} \in \mathbb{C}^{m \times n}$ and $\hat{R} \in \mathbb{C}^{n \times n}$ with real $r_{ii} > 0$.
4. [TB97, p. 48] The orthonormal basis $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n\}$ generated via the Gram–Schmidt orthogonalization process has the property

$$\text{Span}(\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k\}) = \text{Span}(\{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_k\}),$$

for $k = 1, 2, \dots, n$.

5. [TB97, p. 51]

Algorithm 1: Classical Gram–Schmidt Orthogonalization:

input: a basis $\{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n\}$ for a subspace S
 output: an orthonormal basis $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n\}$ for S
 for $j = 1 : n$
 $\mathbf{u}_j := \mathbf{a}_j$
 for $i = 1 : j - 1$
 $r_{ij} := \langle \mathbf{a}_j, \mathbf{u}_i \rangle$
 $\mathbf{u}_j := \mathbf{u}_j - r_{ij}\mathbf{u}_i$
 end
 $r_{jj} := \|\mathbf{u}_j\|$
 $\mathbf{u}_j := \mathbf{u}_j / r_{jj}$
 end

6. [TB97, p. 58]

Algorithm 2: Modified Gram–Schmidt Orthogonalization

input: a basis $\{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n\}$ for a subspace S
 output: an orthonormal basis $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n\}$ for S
 $\mathbf{w}_i := \mathbf{a}_i, i = 1 : n$
 for $i = 1 : n$
 $r_{ii} := \|\mathbf{w}_i\|$
 $\mathbf{u}_i := \mathbf{w}_i / r_{ii}$
 for $j = i + 1 : n$
 $r_{ij} := \langle \mathbf{w}_j, \mathbf{u}_i \rangle$
 $\mathbf{w}_j := \mathbf{w}_j - r_{ij}\mathbf{u}_i$
 end
 end

7. [Mey00, p. 315] If exact arithmetic is used, then Algorithms 1 and 2 generate the same orthonormal basis $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n\}$ and the same r_{ij} , for $j \geq i$.
8. [GV96, pp. 230–232] If $A = [\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n] \in \mathbb{C}^{m \times n}$ ($m \geq n$) is of full rank n , then the classic or modified Gram–Schmidt process leads to a reduced QR factorization $A = \hat{Q}\hat{R}$, with $\hat{Q} = [\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n]$ and $\hat{R}_{ij} = r_{ij}$, for $j \geq i$, and $\hat{R}_{ij} = 0$, for $j < i$.
9. [GV96, p. 232] The costs of Algorithm 1 and Algorithm 2 are both $2mn^2$ flops when applied to compute a reduced QR factorization of a matrix $A \in \mathbb{R}^{m \times n}$.
10. [Mey00, p. 317 and p. 349] For the QR factorization, Algorithm 1 and Algorithm 2 are not numerically stable. However, Algorithm 2 often yields better numerical results than Algorithm 1.
11. [Mey00, p. 349] Algorithm 2 is numerically stable when it is used to solve least squares problems.
12. (Numerically stable algorithms for computing the QR factorization using Householder reflections and Givens rotations are given in Chapter 38.)
13. [TB97, p. 54] (See also Chapter 38.) If $A = QR$ is a QR factorization of the rank n matrix $A \in \mathbb{C}^{n \times n}$, then the linear system $A\mathbf{x} = \mathbf{b}$ can be solved as follows:

- Compute the factorization $A = QR$.
- Compute the vector $\mathbf{c} = Q^*\mathbf{b}$.
- Solve $R\mathbf{x} = \mathbf{c}$ by performing back substitution.

Examples:

1. Consider the matrix $A = \begin{bmatrix} 1 & 2 \\ 2 & 0 \\ 0 & 2 \end{bmatrix}$.

- A has a (full) QR factorization $A = QR$:

$$\begin{bmatrix} 1 & 2 \\ 2 & 0 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{5}} & \frac{4}{3\sqrt{5}} & -\frac{2}{3} \\ \frac{2}{\sqrt{5}} & -\frac{2}{3\sqrt{5}} & \frac{1}{3} \\ 0 & \frac{\sqrt{5}}{3} & \frac{2}{3} \end{bmatrix} \begin{bmatrix} \sqrt{5} & \frac{2}{\sqrt{5}} \\ 0 & \frac{6}{\sqrt{5}} \\ 0 & 0 \end{bmatrix}.$$

- A has a reduced QR factorization $A = \hat{Q}\hat{R}$:

$$\begin{bmatrix} 1 & 2 \\ 2 & 0 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{5}} & \frac{4}{3\sqrt{5}} \\ \frac{2}{\sqrt{5}} & -\frac{2}{3\sqrt{5}} \\ 0 & \frac{\sqrt{5}}{3} \end{bmatrix} \begin{bmatrix} \sqrt{5} & \frac{2}{\sqrt{5}} \\ 0 & \frac{6}{\sqrt{5}} \end{bmatrix}.$$

2. Consider the matrix $A = \begin{bmatrix} 3 & 1 & -2 \\ 3 & -4 & 1 \\ 3 & -4 & -1 \\ 3 & 1 & 0 \end{bmatrix}$. Using the classic or modified Gram–Schmidt process

gives the following reduced QR factorization:

$$\begin{bmatrix} 3 & 1 & -2 \\ 3 & -4 & 1 \\ 3 & -4 & -1 \\ 3 & 1 & 0 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} 6 & -3 & -1 \\ 0 & 5 & -1 \\ 0 & 0 & 2 \end{bmatrix}.$$

5.6 Singular Value Decomposition

Definitions:

A **singular value decomposition (SVD)** of a matrix $A \in \mathbb{C}^{m \times n}$ is a factorization

$$A = U\Sigma V^*, \quad \Sigma = \text{diag}(\sigma_1, \sigma_2, \dots, \sigma_p) \in \mathbb{R}^{m \times n}, \quad p = \min\{m, n\},$$

where $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_p \geq 0$ and both $U = [\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_m] \in \mathbb{C}^{m \times m}$ and $V = [\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n] \in \mathbb{C}^{n \times n}$ are unitary. The diagonal entries of Σ are called the **singular values** of A . The columns of U are called **left singular vectors** of A and the columns of V are called **right singular vectors** of A .

Let $A \in \mathbb{C}^{m \times n}$ with rank $r \leq p = \min\{m, n\}$. A **reduced singular value decomposition (reduced SVD)** of A is a factorization

$$A = \hat{U}\hat{\Sigma}\hat{V}^*, \quad \hat{\Sigma} = \text{diag}(\sigma_1, \sigma_2, \dots, \sigma_r) \in \mathbb{R}^{r \times r},$$

where $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r > 0$ and the columns of $\hat{U} = [\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_r] \in \mathbb{C}^{m \times r}$ and the columns of $\hat{V} = [\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r] \in \mathbb{C}^{n \times r}$ are both orthonormal.

(See §8.4 and §3.7 for more information on singular value decomposition.)

Facts:

All the following facts except those with a specific reference can be found in [TB97, pp. 25–37].

1. Every $A \in \mathbb{C}^{m \times n}$ has a singular value decomposition $A = U\Sigma V^*$. If $A \in \mathbb{R}^{m \times n}$, then U and V may be taken to be real.
2. The singular values of a matrix are uniquely determined.
3. If $A \in \mathbb{C}^{m \times n}$ has a singular value decomposition $A = U\Sigma V^*$, then

$$A\mathbf{v}_j = \sigma_j \mathbf{u}_j, \quad A^* \mathbf{u}_j = \sigma_j \mathbf{v}_j, \quad \mathbf{u}_j^* A \mathbf{v}_j = \sigma_j,$$

for $j = 1, 2, \dots, p = \min\{m, n\}$.

4. If $U\Sigma V^*$ is a singular value decomposition of A , then $V\Sigma^T U^*$ is a singular value decomposition of A^* .
5. If $A \in \mathbb{C}^{m \times n}$ has r nonzero singular values, then
 - $\text{rank}(A) = r$.
 - $A = \sum_{j=1}^r \sigma_j \mathbf{u}_j \mathbf{v}_j^*$.
 - $\ker(A) = \text{span}\{\mathbf{v}_{r+1}, \dots, \mathbf{v}_n\}$.
 - $\text{range}(A) = \text{span}\{\mathbf{u}_1, \dots, \mathbf{u}_r\}$.
6. Any $A \in \mathbb{C}^{m \times n}$ of rank $r \leq p = \min\{m, n\}$ has a **reduced singular value decomposition**,

$$A = \hat{U} \hat{\Sigma} \hat{V}^*, \quad \hat{\Sigma} = \text{diag}(\sigma_1, \sigma_2, \dots, \sigma_r) \in \mathbb{R}^{r \times r},$$

where $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r > 0$ and the columns of $\hat{U} = [\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_r] \in \mathbb{C}^{m \times r}$ and the columns of $\hat{V} = [\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r] \in \mathbb{C}^{n \times r}$ are both orthonormal. If $A \in \mathbb{R}^{m \times n}$, then $\hat{U}$ and $\hat{V}$ may be taken to be real.

7. If $\text{rank}(A) = r$, then A has r nonzero singular values.
8. The nonzero singular values of A are the square roots of the nonzero eigenvalues of A^*A or AA^* .
9. [HJ85, p. 414] If $U\Sigma V^*$ is a singular value decomposition of A , then the columns of V are eigenvectors of A^*A ; the columns of U are eigenvectors of AA^* .
10. [HJ85, p. 418] Let $A \in \mathbb{C}^{m \times n}$ and $p = \min\{m, n\}$. Define

$$G = \begin{bmatrix} 0 & A \\ A^* & 0 \end{bmatrix} \in \mathbb{C}^{(m+n) \times (m+n)}.$$

If the singular values of A are $\sigma_1, \dots, \sigma_p$, then the eigenvalues of G are $\sigma_1, \dots, \sigma_p, -\sigma_1, \dots, -\sigma_p$ and additional $|n - m|$ zeros.

11. If $A \in \mathbb{C}^{n \times n}$ is Hermitian with eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$, then the singular values of A are $|\lambda_1|, |\lambda_2|, \dots, |\lambda_n|$.
12. For $A \in \mathbb{C}^{n \times n}$, $|\det A| = \sigma_1 \sigma_2 \dots \sigma_n$.
13. [Aut15; Sch07] (**Eckart–Young Low Rank Approximation Theorem**)

Let $A = U\Sigma V^*$ be an SVD of $A \in \mathbb{C}^{m \times n}$ and $r = \text{rank}(A)$. For $k < r$, define $A_k = \sum_{j=1}^k \sigma_j \mathbf{u}_j \mathbf{v}_j^*$. Then

- $\|A - A_k\|_2 = \min_{\text{rank}(B) \leq k} \|A - B\|_2 = \sigma_{k+1}$;
- $\|A - A_k\|_F = \min_{\text{rank}(B) \leq k} \|A - B\|_F = \sqrt{\sum_{j=k+1}^r \sigma_j^2}$,

where $\|M\|_2 = \max_{\|x\|_2=1} \|Mx\|_2$ and $\|M\|_F = \sqrt{\sum_{i=1}^m \sum_{j=1}^n m_{ij}^2}$ are the 2-norm and Frobenius norm of matrix M , respectively. (See Chapter 37 for more information on matrix norms.)

Examples:

Consider the matrices $A = \begin{bmatrix} 1 & 2 \\ 2 & 0 \\ 0 & 2 \end{bmatrix}$ and $B = A^T = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 0 & 2 \end{bmatrix}$.

1. The eigenvalues of $A^T A = \begin{bmatrix} 5 & 2 \\ 2 & 8 \end{bmatrix}$ are 9 and 4. So, the singular values of A are 3 and 2.
2. Normalized eigenvectors for $A^T A$ are $\mathbf{v}_1 = \begin{bmatrix} \frac{1}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} \end{bmatrix}$ and $\mathbf{v}_2 = \begin{bmatrix} \frac{2}{\sqrt{5}} \\ -\frac{1}{\sqrt{5}} \end{bmatrix}$.
3. $\mathbf{u}_1 = \frac{1}{3} A \mathbf{v}_1 = \begin{bmatrix} \frac{\sqrt{5}}{3} \\ \frac{2}{3\sqrt{5}} \\ \frac{4}{3\sqrt{5}} \end{bmatrix}$ and $\mathbf{u}_2 = \frac{1}{2} A \mathbf{v}_2 = \begin{bmatrix} 0 \\ \frac{2}{\sqrt{5}} \\ -\frac{1}{\sqrt{5}} \end{bmatrix}$. Application of the Gram–Schmidt process to $\mathbf{u}_1, \mathbf{u}_2$, and $\mathbf{e}_1$ produces $\mathbf{u}_3 = \begin{bmatrix} \frac{2}{3} \\ -\frac{1}{3} \\ -\frac{2}{3} \end{bmatrix}$.

4. A has the singular value decomposition $A = U \Sigma V^T$, where

$$U = \frac{1}{3\sqrt{5}} \begin{bmatrix} 5 & 0 & 2\sqrt{5} \\ 2 & 6 & -\sqrt{5} \\ 4 & -3 & -2\sqrt{5} \end{bmatrix}, \quad \Sigma = \begin{bmatrix} 3 & 0 \\ 0 & 2 \\ 0 & 0 \end{bmatrix}, \quad V = \frac{1}{\sqrt{5}} \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}.$$

5. A has the reduced singular value decomposition $A = \hat{U} \hat{\Sigma} \hat{V}^T$, where

$$\hat{U} = \frac{1}{3\sqrt{5}} \begin{bmatrix} 5 & 0 \\ 2 & 6 \\ 4 & -3 \end{bmatrix}, \quad \hat{\Sigma} = \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}, \quad \hat{V} = \frac{1}{\sqrt{5}} \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}.$$

6. B has the singular value decomposition $B = U_B \Sigma_B V_B^T$, where

$$U_B = V_A = \frac{1}{\sqrt{5}} \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}, \quad \Sigma_B = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \end{bmatrix}, \quad V_B = U_A = \frac{1}{3\sqrt{5}} \begin{bmatrix} 5 & 0 & 2\sqrt{5} \\ 2 & 6 & -\sqrt{5} \\ 4 & -3 & -2\sqrt{5} \end{bmatrix}.$$

($U_A = U$ and $V_A = V$ for A were given in Example 4.)

5.7 Pseudo-Inverse

Definitions:

A **Moore–Penrose pseudo-inverse** of a matrix $A \in \mathbb{C}^{m \times n}$ is a matrix $A^\dagger \in \mathbb{C}^{n \times m}$ that satisfies the following four **Penrose** conditions:

$$AA^\dagger A = A; \quad A^\dagger AA^\dagger = A^\dagger; \quad (AA^\dagger)^* = AA^\dagger; \quad (A^\dagger A)^* = A^\dagger A.$$

Facts:

All the following facts except those with a specific reference can be found in [Gra83, pp. 105–141].

1. Every $A \in \mathbb{C}^{m \times n}$ has a unique pseudo-inverse $A^\dagger$. If $A \in \mathbb{R}^{m \times n}$, then $A^\dagger$ is real.

2. [LH95, p. 38] If $A \in \mathbb{C}^{m \times n}$ of rank $r \leq \min\{m, n\}$ has an SVD $A = U\Sigma V^*$, then its pseudo-inverse is $A^\dagger = V\Sigma^\dagger U^*$, where

$$\Sigma^\dagger = \text{diag}(1/\sigma_1, \dots, 1/\sigma_r, 0, \dots, 0) \in \mathbb{R}^{n \times m}.$$

3. $\mathbf{0}_{mn}^\dagger = \mathbf{0}_{nm}$ and $J_{mn}^\dagger = \frac{1}{mn} J_{nm}$, where $\mathbf{0}_{mn} \in \mathbb{C}^{m \times n}$ is the all 0s matrix and $J_{mn} \in \mathbb{C}^{m \times n}$ is the all 1s matrix.
4. $(A^\dagger)^* = (A^*)^\dagger$; $(A^\dagger)^\dagger = A$.
5. If A is a nonsingular square matrix, then $A^\dagger = A^{-1}$.
6. If U has orthonormal columns or orthonormal rows, then $U^\dagger = U^*$.
7. If $A = A^*$ and $A = A^2$, then $A^\dagger = A$.
8. $A^\dagger = A^*$ if and only if A^*A is idempotent.
9. If $A = A^*$, then $AA^\dagger = A^\dagger A$.
10. If $U \in \mathbb{C}^{m \times n}$ is of rank n and satisfies $U^\dagger = U^*$, then U has orthonormal columns.
11. If $U \in \mathbb{C}^{m \times m}$ and $V \in \mathbb{C}^{n \times n}$ are unitary matrices, then $(UAV)^\dagger = V^*A^\dagger U^*$.
12. If $A \in \mathbb{C}^{m \times n}$ ($m \geq n$) has full rank n , then $A^\dagger = (A^*A)^{-1}A^*$.
13. If $A \in \mathbb{C}^{m \times n}$ ($m \leq n$) has full rank m , then $A^\dagger = A^*(AA^*)^{-1}$.
14. Let $A \in \mathbb{C}^{m \times n}$. Then
 - $A^\dagger A$, $AA^\dagger$, $I_n - A^\dagger A$, and $I_m - AA^\dagger$ are orthogonal projections.
 - $\text{rank}(A) = \text{rank}(A^\dagger) = \text{rank}(AA^\dagger) = \text{rank}(A^\dagger A)$.
 - $\text{rank}(I_n - A^\dagger A) = n - \text{rank}(A)$.
 - $\text{rank}(I_m - AA^\dagger) = m - \text{rank}(A)$.
15. If $A = A_1 + A_2 + \dots + A_k$, $A_i^*A_j = 0$, and $A_iA_j^* = 0$, for all $i, j = 1, \dots, k$, $i \neq j$, then $A^\dagger = A_1^\dagger + A_2^\dagger + \dots + A_k^\dagger$.
16. If A is an $m \times r$ matrix of rank r and B is an $r \times n$ matrix of rank r , then $(AB)^\dagger = B^\dagger A^\dagger$.
17. $(A^*A)^\dagger = A^\dagger(A^*)^\dagger$; $(AA^*)^\dagger = (A^*)^\dagger A^\dagger$.
18. [Gre66] Each one of the following conditions is necessary and sufficient for $(AB)^\dagger = B^\dagger A^\dagger$:
 - $\text{range}(BB^*A^*) \subseteq \text{range}(A^*)$ and $\text{range}(A^*AB) \subseteq \text{range}(B)$.
 - $A^\dagger ABB^*$ and $A^*ABB^\dagger$ are both Hermitian matrices.
 - $A^\dagger ABB^*A^* = BB^*A^*$ and $BB^\dagger A^*AB = A^*AB$.
 - $A^\dagger ABB^*A^*ABB^\dagger = BB^*A^*$.
 - $A^\dagger AB = B(AB)^\dagger AB$ and $BB^\dagger A^* = A^*AB(AB)^\dagger$.
19. Let $A \in \mathbb{C}^{m \times n}$ and $\mathbf{b} \in \mathbb{C}^m$. Then the system of equations $A\mathbf{x} = \mathbf{b}$ is consistent if and only if $AA^\dagger\mathbf{b} = \mathbf{b}$. Moreover, if $A\mathbf{x} = \mathbf{b}$ is consistent, then any solution to the system can be expressed as $\mathbf{x} = A^\dagger\mathbf{b} + (I_n - A^\dagger A)\mathbf{y}$ for some $\mathbf{y} \in \mathbb{C}^n$.

Examples:

1. The pseudo-inverse of the matrix $A = \begin{bmatrix} 1 & 2 \\ 2 & 0 \\ 0 & 2 \end{bmatrix}$ is $A^\dagger = \frac{1}{18} \begin{bmatrix} 2 & 8 & -2 \\ 4 & -2 & 5 \end{bmatrix}$.
2. $(AB)^\dagger = B^\dagger A^\dagger$ generally does not hold. For example, if

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix},$$

then

$$(AB)^\dagger = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}^\dagger = \frac{1}{2} \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}.$$

However,

$$B^\dagger A^\dagger = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}.$$

5.8 Least Squares Problems

Definitions:

Given $A \in F^{m \times n}$ ($F = \mathbb{R}$ or $\mathbb{C}$), $m \geq n$, and $\mathbf{b} \in F^m$, the **least squares problem** is to find an $\mathbf{x}_0 \in F^n$ such that $\|\mathbf{b} - A\mathbf{x}\|$ is minimized: $\|\mathbf{b} - A\mathbf{x}_0\| = \min_{\mathbf{x} \in F^n} \|\mathbf{b} - A\mathbf{x}\|$.

- Such an $\mathbf{x}_0$ is called a **solution** to the least squares problem or a **least squares solution** to the linear system $A\mathbf{x} = \mathbf{b}$.
- The vector $\mathbf{r} = \mathbf{b} - A\mathbf{x} \in F^m$ is called the **residual**.
- If $\text{rank}(A) = n$, then the least squares problem is called the **full rank least squares problem**.
- If $\text{rank}(A) < n$, then the least squares problem is called the **rank-deficient least squares problem**.

The system $A^*A\mathbf{x} = A^*\mathbf{b}$ is called the **normal equation** for the least squares problem.

(See Chapter 39 for more information on least squares problems.)

Facts:

1. [Mey00, p. 439] Let $A \in F^{m \times n}$ ($F = \mathbb{R}$ or $\mathbb{C}$, $m \geq n$) and $\mathbf{b} \in F^m$ be given. Then the following statements are equivalent:
 - $\mathbf{x}_0$ is a solution for the least squares problem.
 - $\min_{\mathbf{x} \in F^n} \|\mathbf{b} - A\mathbf{x}\| = \|\mathbf{b} - A\mathbf{x}_0\|$.
 - $A\mathbf{x}_0 = P\mathbf{b}$, where P is the orthogonal projection onto $\text{range}(A)$.
 - $A^*\mathbf{r}_0 = \mathbf{0}$, where $\mathbf{r}_0 = \mathbf{b} - A\mathbf{x}_0$.
 - $A^*A\mathbf{x}_0 = A^*\mathbf{b}$.
 - $\mathbf{x}_0 = A^\dagger\mathbf{b} + \mathbf{y}_0$ for some $\mathbf{y}_0 \in \ker(A)$.
2. [LH95, p. 36] If $A \in F^{m \times n}$ ($F = \mathbb{R}$ or $\mathbb{C}$, $m \geq n$) and $\text{rank}(A) = r \leq n$, then $\mathbf{x}_0 = A^\dagger\mathbf{b}$ is the unique solution of minimum length for the least squares problem.
3. [TB97, p. 81] If $A \in F^{m \times n}$ ($F = \mathbb{R}$ or $\mathbb{C}$, $m \geq n$) has full rank, then $\mathbf{x}_0 = A^\dagger\mathbf{b} = (A^*A)^{-1}A^*\mathbf{b}$ is the unique solution for the least squares problem.
4. [TB97, p. 83]

Algorithm 3: Solving Full Rank Least Squares via QR Factorization

input: matrix $A \in F^{m \times n}$ ($F = \mathbb{R}$ or $\mathbb{C}$, $m \geq n$) with full rank n and vector $\mathbf{b} \in F^m$

output: solution $\mathbf{x}_0$ for $\min_{\mathbf{x} \in F^n} \|\mathbf{b} - A\mathbf{x}\|$

compute the reduced QR factorization $A = \hat{Q}\hat{R}$;

compute the vector $\mathbf{c} = \hat{Q}^*\mathbf{b}$;

solve $\hat{R}\mathbf{x}_0 = \mathbf{c}$ using back substitution.

5. [TB97, p. 84]

Algorithm 4: Solving Full Rank Least Squares via SVD

input: matrix $A \in F^{m \times n}$ ($F = \mathbb{R}$ or $\mathbb{C}$, $m \geq n$) with full rank n and vector $\mathbf{b} \in F^m$

output : solution $\mathbf{x}_0$ for $\min_{\mathbf{x} \in F^n} \|\mathbf{b} - A\mathbf{x}\|$

compute the reduced SVD $A = \hat{U} \hat{\Sigma} \hat{V}^*$ with $\hat{\Sigma} = \text{diag}(\sigma_1, \sigma_2, \dots, \sigma_n)$;

compute the vector $\mathbf{c} = \hat{U}^* \mathbf{b}$;

compute the vector $\mathbf{y}$: $y_i = c_i / \sigma_i$, $i = 1, 2, \dots, n$;

compute $\mathbf{x}_0 = \hat{V} \mathbf{y}$.

6. [TB97, p. 82]

Algorithm 5: Solving Full Rank Least Squares via Normal Equations

input: matrix $A \in F^{m \times n}$ ($F = \mathbb{R}$ or $\mathbb{C}$, $m \geq n$) with full rank n and vector $\mathbf{b} \in F^m$

output : solution $\mathbf{x}_0$ for $\min_{\mathbf{x} \in F^n} \|\mathbf{b} - A\mathbf{x}\|$

compute the matrix $A^* A$ and the vector $\mathbf{c} = A^* \mathbf{b}$;

solve the system $A^* A \mathbf{x}_0 = \mathbf{c}$ via the Cholesky factorization.

Examples:

1. Consider the inconsistent linear system $A\mathbf{x} = \mathbf{b}$, where

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 0 \\ 0 & 2 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}.$$

Then the normal equations are given by $A^T A \mathbf{x} = A^T \mathbf{b}$, where

$$A^T A = \begin{bmatrix} 5 & 2 \\ 2 & 8 \end{bmatrix} \quad \text{and} \quad A^T \mathbf{b} = \begin{bmatrix} 5 \\ 8 \end{bmatrix}.$$

A least squares solution to the system $A\mathbf{x} = \mathbf{b}$ can be obtained via solving the normal equations:

$$\mathbf{x}_0 = (A^T A)^{-1} A^T \mathbf{b} = A^\dagger \mathbf{b} = \begin{bmatrix} 2/3 \\ 5/6 \end{bmatrix}.$$

2. We use Algorithm 3 to find a least squares solution of the system $A\mathbf{x} = \mathbf{b}$ given in Example 1. The reduced QR factorization $A = \hat{Q} \hat{R}$ found in Example 1 in Section 5.5 gives

$$\hat{Q}^T \mathbf{b} = \begin{bmatrix} \frac{1}{\sqrt{5}} & \frac{4}{3\sqrt{5}} \\ \frac{2}{\sqrt{5}} & -\frac{2}{3\sqrt{5}} \\ 0 & \frac{\sqrt{5}}{3} \end{bmatrix}^T \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} \sqrt{5} \\ \sqrt{5} \end{bmatrix}.$$

Now solving $\hat{R}\mathbf{x} = [\sqrt{5}, \sqrt{5}]^T$ gives the least squares solution $\mathbf{x}_0 = [2/3, 5/6]^T$.

3. We use Algorithm 4 to solve the same problem given in Example 1. Using the reduced singular value decomposition $A = \hat{U} \hat{\Sigma} \hat{V}^T$ obtained in Example 5, Section 5.6, we have

$$\mathbf{c} = \hat{U}^T \mathbf{b} = \frac{1}{3\sqrt{5}} \begin{bmatrix} 5 & 0 \\ 2 & 6 \\ 4 & -3 \end{bmatrix}^T \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} \frac{7}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} \end{bmatrix}.$$

Now we compute $\mathbf{y} = [y_1, y_2]^T$:

$$y_1 = c_1/\sigma_1 = \frac{7}{3\sqrt{5}} \quad \text{and} \quad y_2 = c_2/\sigma_2 = \frac{1}{2\sqrt{5}}.$$

Finally, the least squares solution is obtained via

$$\mathbf{x}_0 = \hat{\mathbf{V}}\mathbf{y} = \frac{1}{\sqrt{5}} \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} \frac{7}{3\sqrt{5}} \\ \frac{1}{2\sqrt{5}} \end{bmatrix} = \begin{bmatrix} 2/3 \\ 5/6 \end{bmatrix}.$$

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Matrices with Special Properties

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6

Canonical Forms

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A *canonical form* of a matrix is a special form with the properties that every matrix is associated to a matrix in that form (the canonical form of the matrix), it is unique or essentially unique (typically up to some type of permutation), and it has a particularly simple form (or a form well suited to a specific purpose). A canonical form partitions the set matrices in $F^{m \times n}$ into sets of matrices each having the same canonical form, and that canonical form matrix serves as the representative. The canonical form of a given matrix can provide important information about the matrix. For example, reduced row echelon form (RREF) is a canonical form that is useful in solving systems of linear equations; RREF partitions $F^{m \times n}$ into sets of row equivalent matrices.

The previous definition of a canonical form is far more general than the canonical forms discussed in this chapter. Here all matrices are square, and every matrix is similar to its canonical form. This chapter discusses the two most important canonical forms for square matrices over fields, the Jordan canonical form (and its real version) and (two versions of) the rational canonical form. These canonical forms capture the eigenstructure of a matrix and play important roles in many areas, for example, in matrix functions, Chapter 11, and in differential equations, Chapter 55. These canonical forms partition $F^{n \times n}$ into similarity classes.

The Jordan canonical form is most often used when all eigenvalues of the matrix $A \in F^{n \times n}$ lie in the field F , such as when the field is algebraically closed (e.g., $\mathbb{C}$), or when the field is $\mathbb{R}$; otherwise the rational canonical form is used (e.g., for $\mathbb{Q}$). The Smith normal form is a canonical form for square matrices over principal ideal domains (see Chapter 23); it is discussed here only as it pertains to the computation of the rational canonical form. If any one of these canonical forms is known, it is straightforward to determine the others (perhaps in the algebraic closure of the field F). Details are given in the sections on rational canonical form.

Results about each type of canonical form are presented in the section on that canonical form, which facilitates locating a result, but obscures the connections underlying the derivations of the results. The facts about all of the canonical forms discussed in this section can be derived from results about modules over a principal ideal domain; such a module-theoretic treatment is typically presented in abstract algebra texts, such as [DF04, Chap. 12].

None of the canonical forms discussed in this chapter is a continuous function of the entries of a matrix and, thus, the computation of such a canonical form is inherently unstable in finite precision arithmetic. (For information about perturbation theory of eigenvalues see Chapter 15; for information specifically about numerical computation of the Jordan canonical form, see [GV96, Chapter 7.6.5].)

6.1 Generalized Eigenvectors

The reader is advised to consult Section 4.3 for information about eigenvalues and eigenvectors. In this section and the next, F is taken to be an algebraically closed field to ensure that an $n \times n$ matrix has n eigenvalues, but many of the results could be rephrased for a matrix that has all its eigenvalues in F , without the assumption that F is algebraically closed. The real versions of the definitions and results are presented in Section 6.3.

Definitions:

Let F be an algebraically closed field (e.g., $\mathbb{C}$), let $A \in F^{n \times n}$, let $\mu_1, \dots, \mu_r$ be the distinct eigenvalues of A , and let λ be any eigenvalue of A .

For k a nonnegative integer, the k -**eigenspace** of A at λ , denoted $N_\lambda^k(A)$, is $\ker(A - \lambda I)^k$.

The **index** of A at λ , denoted $v_\lambda(A)$, is the smallest integer k such that $N_\lambda^k(A) = N_\lambda^{k+1}(A)$. When λ and A are clear from the context, $v_\lambda(A)$ will be abbreviated to v , and $v_{\mu_i}(A)$ to v_i .

The **generalized eigenspace** of A at λ is the set $N_\lambda^{v_\lambda}(A)$, where v is the index of A at λ .

The vector $\mathbf{x} \in F^n$ is a **generalized eigenvector** of A for λ if $\mathbf{x} \neq \mathbf{0}$ and $\mathbf{x} \in N_\lambda^{v_\lambda}(A)$.

Let V be a finite dimensional vector space over F , and let T be a linear operator on V . The definitions of k -**eigenspace** of T , **index**, and **generalized eigenspace** of T are analogous.

Facts:

Facts requiring proof for which no specific reference is given can be found in [HJ85, Chapter 3] or [Mey00, Chapter 7.8].

Notation: F is an algebraically closed field, $A \in F^{n \times n}$, V is an n dimensional vector space over F , $T \in L(V, V)$, $\mu_1, \dots, \mu_r$ are the distinct eigenvalues of A or T , and $\lambda = \mu_i$ for some $i \in \{1, \dots, r\}$.

1. An eigenvector for eigenvalue λ is a generalized eigenvector for λ , but the converse is not necessarily true.
2. The eigenspace for λ is the 1-eigenspace, i.e., $E_\lambda(A) = N_\lambda^1(A)$.
3. Every k -eigenspace is invariant under multiplication by A .
4. The dimension of the generalized eigenspace of A at λ is the algebraic multiplicity of λ , i.e., $\dim N_{\mu_i}^{v_i}(A) = \alpha_A(\mu_i)$.
5. A is diagonalizable if and only if $v_i = 1$ for $i = 1, \dots, r$.
6. F^n is the vector space direct sum of the generalized eigenspaces, i.e.,

$$F^n = N_{\mu_1}^{v_1}(A) \oplus \dots \oplus N_{\mu_r}^{v_r}(A).$$

This is a special case of the Primary Decomposition Theorem (Fact 12 in Section 6.4).

7. Facts 1 to 6 remain true when the matrix A is replaced by the linear operator T .
8. If $\hat{T}$ denotes T restricted to $N_{\mu_i}^{v_i}(T)$, then the characteristic polynomial of $\hat{T}$ is $p_{\hat{T}}(x) = (x - \mu_i)^{\alpha(\mu_i)}$. In particular, $\hat{T} - \mu_i I$ is nilpotent.

Examples:

$$1. \text{ Let } A = \begin{bmatrix} 65 & 18 & -21 & 4 \\ -201 & -56 & 63 & -12 \\ 67 & 18 & -23 & 4 \\ 134 & 36 & -42 & 6 \end{bmatrix} \in \mathbb{C}^{4 \times 4}, p_A(x) = x^4 + 8x^3 + 24x^2 + 32x + 16 = (x + 2)^4,$$

so the only eigenvalue of A is -2 with algebraic multiplicity 4. The reduced row echelon form of

$$A + 2I \text{ is } \begin{bmatrix} 1 & \frac{18}{67} & -\frac{21}{67} & \frac{4}{67} \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \text{ so } N_{-2}^1(A) = \text{Span} \left(\begin{bmatrix} -18 \\ 67 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 21 \\ 0 \\ 67 \\ 0 \end{bmatrix}, \begin{bmatrix} -4 \\ 0 \\ 0 \\ 67 \end{bmatrix} \right).$$

$(A + 2I)^2 = 0$, so $N_{-2}^2(A) = \mathbb{C}^4$. Any vector not in $N_{-2}^1(A)$, e.g., $\mathbf{e}_1 = [1, 0, 0, 0]^T$, is a generalized eigenvector for -2 that is not an eigenvector for -2 .

6.2 Jordan Canonical Form

The Jordan canonical form is perhaps the single most important and widely used similarity-based canonical form for (square) matrices.

Definitions:

Let F be an algebraically closed field (e.g., $\mathbb{C}$), and let $A \in F^{n \times n}$. (The real versions of the definitions and results are presented in Section 6.3.)

For $\lambda \in F$ and positive integer k , the **Jordan block** of size k with eigenvalue λ is the $k \times k$ matrix having every diagonal entry equal to λ , every first superdiagonal entry equal to 1, and every other entry equal to 0, i.e.,

$$J_k(\lambda) = \begin{bmatrix} \lambda & 1 & 0 & \cdots & 0 \\ 0 & \lambda & 1 & & 0 \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & \lambda & 1 \\ 0 & \cdots & 0 & 0 & \lambda \end{bmatrix}.$$

A **Jordan matrix** (or a **matrix in Jordan canonical form**) is a block diagonal matrix having Jordan blocks as the diagonal blocks, i.e., a matrix of the form $J_{k_1}(\lambda_1) \oplus \cdots \oplus J_{k_t}(\lambda_t)$ for some positive integers $t, k_1, \dots, k_t$ and some $\lambda_1, \dots, \lambda_t \in F$. (Note: the λ_i need not be distinct.)

A **Jordan canonical form** of matrix A , denoted J_A or $\text{JCF}(A)$, is a Jordan matrix that is similar to A . It is conventional to group the blocks for the same eigenvalue together and to order the Jordan blocks with the same eigenvalue in nonincreasing size order.

The **Jordan invariants** of A are the following parameters:

- The set of distinct eigenvalues of A .
- For each eigenvalue λ , the number b_λ and sizes $p_1, \dots, p_{b_\lambda}$ of the Jordan blocks with eigenvalue λ in a Jordan canonical form of A .

The **total number of Jordan blocks** in a Jordan canonical form of A is $\sum b_\mu$, where the sum is taken over all distinct eigenvalues μ .

If $J_A = C^{-1}AC$, then the ordered set of columns of C is called a **Jordan basis** for A .

Let $\mathbf{x}$ be an eigenvector for eigenvalue λ of A . If $\mathbf{x} \in \text{range}(A - \lambda I)^h - \text{range}(A - \lambda I)^{h+1}$. Then h is called the **depth** of $\mathbf{x}$.

Let $\mathbf{x}$ be an eigenvector of depth h for eigenvalue λ of A . A **Jordan chain** above $\mathbf{x}$ is a sequence of vectors $\mathbf{x}_0 = \mathbf{x}, \mathbf{x}_1, \dots, \mathbf{x}_h$ satisfying $\mathbf{x}_i = (A - \lambda I)\mathbf{x}_{i+1}$ for $i = 0, \dots, h-1$.

Let V be a finite dimensional vector space over F , and let T be a linear operator on V .

A **Jordan basis** for T is an ordered basis $\mathcal{B}$ of V , with respect to which the matrix ${}_B[T]_B$ of T is a Jordan matrix. In this case, ${}_B[T]_B$ is a Jordan canonical form of T , denoted $\text{JCF}(T)$ or J_T , and the **Jordan invariants** of T are the Jordan invariants of $\text{JCF}(T) = {}_B[T]_B$.

Facts:

Facts requiring proof for which no specific reference is given can be found in [HJ85, Chapter 3] or [Mey00, Chapter 7.8].

Notation: F is an algebraically closed field, $A, B \in F^{n \times n}$, and λ is an eigenvalue of A .

1. A has a Jordan canonical form J_A , and J_A is unique up to permutation of the Jordan blocks. In particular, the Jordan invariants of A are uniquely determined by A .
2. A, B are similar if and only if they have the same Jordan invariants.
3. The Jordan invariants and, hence, the Jordan canonical form of A can be found from the eigenvalues and the ranks of powers of $A - \lambda I$. Specifically, the number of Jordan blocks of size k in J_A with eigenvalue λ is

$$\text{rank}(A - \lambda I)^{k-1} + \text{rank}(A - \lambda I)^{k+1} - 2 \text{rank}(A - \lambda I)^k.$$

4. The total number of Jordan blocks in a Jordan canonical form of A is the maximal number of linearly independent eigenvectors of A .
5. The number b_λ of Jordan blocks with eigenvalue λ in J_A equals the geometric multiplicity $\gamma_A(\lambda)$ of λ . A is nonderogatory if and only if for each eigenvalue λ of A , J_A has exactly one block with λ .
6. The size of the largest Jordan block with eigenvalue λ equals the multiplicity of λ as a root of the minimal polynomial $q_A(x)$ of A .
7. The size of the largest Jordan block with eigenvalue λ equals the size of the index $v_\lambda(A)$ of A at λ .
8. The sum of the sizes of all the Jordan blocks with eigenvalue λ in J_A (i.e., the number of times λ appears on the diagonal of the Jordan canonical form) equals the algebraic multiplicity $\alpha_A(\lambda)$ of λ .
9. Knowledge of both the characteristic and minimal polynomials suffices to determine the Jordan block sizes for any eigenvalue having algebraic multiplicity at most 3 and, hence, to determine the Jordan canonical form of A if no eigenvalue of A has algebraic multiplicity exceeding 3. This is not necessarily true when the algebraic multiplicity of an eigenvalue is 4 or greater (cf. Example 3 below).
10. Knowledge of the the algebraic multiplicity, geometric multiplicity, and index of an eigenvalue λ suffices to determine the Jordan block sizes for λ if the algebraic multiplicity of λ is at most 6. This is not necessarily true when the algebraic multiplicity of an eigenvalue is 7 or greater (cf. Example 4 below).
11. The following are equivalent:
 - (a) A is similar to a diagonal matrix.
 - (b) The total number of Jordan blocks of A equals n .
 - (c) The size of every Jordan block in a Jordan canonical form J_A of A is 1.
12. If A is real, then nonreal eigenvalues of A occur in conjugate pairs; furthermore, if λ is a nonreal eigenvalue, then each size k Jordan block with eigenvalue λ can be paired with a size k Jordan block for $\bar{\lambda}$.
13. If $A = A_1 \oplus \dots \oplus A_m$, then $J_{A_1} \oplus \dots \oplus J_{A_m}$ is a Jordan canonical form of A .
14. [Mey00, Chapter 7.8] A Jordan basis and Jordan canonical form of A can be constructed by using Algorithm 1.

Algorithm 1: Jordan Basis and Jordan Canonical Form

Input: $A \in F^{n \times n}$, the distinct eigenvalues $\mu_1, \dots, \mu_r$, the indices $v_1, \dots, v_r$.

Output: $C \in F^{n \times n}$ such that $C^{-1}AC = J_A$.

Initially C has no columns.

FOR $i = 1, \dots, r$ % working on eigenvalue μ_i

Step 1: Find a special basis $\mathcal{B}_{\mu_i}$ for $E_{\mu_i}(A)$.

 (a) Initially $\mathcal{B}_{\mu_i}$ has no vectors.

 (b) FOR $k = v_i - 1$ down to 0

 Extend the set of vectors already found to a basis for $\text{range}(A - \mu_i I)^k \cap E_{\mu_i}(A)$.

 (c) Denote the vectors of $\mathcal{B}_{\mu_i}$ by $\mathbf{b}_j$ (ordered as found in step (b)).

Step 2: For each vector $\mathbf{b}_j$ found in Step 1, build a Jordan chain above $\mathbf{b}_j$.

 FOR $j = 1, \dots, \dim \ker(A - \mu_i I)$ % working on $\mathbf{b}_j$

 (a) Solve $(A - \mu_i I)^{h_j} \mathbf{u}_j = \mathbf{b}_j$ for $\mathbf{u}_j$ where h_j is the depth of $\mathbf{b}_j$.

 (b) Insert $(A - \mu_i I)^{h_j} \mathbf{u}_j, (A - \mu_i I)^{h_j-1} \mathbf{u}_j, \dots, (A - \mu_i I) \mathbf{u}_j, \mathbf{u}_j$
 as the next $h + 1$ columns of C .

15. A and its transpose A^T have the same Jordan canonical form (and are, therefore, similar).
16. For a nilpotent matrix, the list of block sizes determines the Jordan canonical form or, equivalently, determines the similarity class. The number of similarity classes of nilpotent matrices of size n is the number of partitions of n .
17. Let J_A be a Jordan matrix, let D be the diagonal matrix having the same diagonal as J_A , and let $N = J_A - D$. Then N is nilpotent.
18. A can be expressed as the sum of a diagonalizable matrix A_D and a nilpotent matrix A_N , where A_D and A_N are polynomials in A (and A_D and A_N commute).
19. Let V be an n -dimensional vector space over F and T be a linear operator on V . Facts 1, 3 to 10, 16, and 18 remain true when matrix A is replaced by linear operator T ; in particular, $\text{JCF}(T)$ exists and is independent (up to permutation of the diagonal Jordan blocks) of the ordered basis of V used to compute it, and the Jordan invariants of T are independent of basis.

Examples:

$$1. J_4(3) = \begin{bmatrix} 3 & 1 & 0 & 0 \\ 0 & 3 & 1 & 0 \\ 0 & 0 & 3 & 1 \\ 0 & 0 & 0 & 3 \end{bmatrix}.$$

2. Let A be the matrix in Example 1 in Section 6.1. $p_A(x) = x^4 + 8x^3 + 24x^2 + 32x + 16 = (x+2)^4$, so the only eigenvalue of A is -2 with algebraic multiplicity 4. From Example 1 in section 6.1, A has 3 linearly independent eigenvectors for eigenvalue -2 , so J_A has 3 Jordan blocks with eigenvalue -2 .

In this case, this is enough information to completely determine that $J_A = \begin{bmatrix} -2 & 1 & 0 & 0 \\ 0 & -2 & 0 & 0 \\ 0 & 0 & -2 & 0 \\ 0 & 0 & 0 & -2 \end{bmatrix}$.

3. The Jordan canonical form of A is not necessarily determined by the characteristic and minimal polynomials of A . For example, the Jordan matrices $A = J_2(0) \oplus J_1(0) \oplus J_1(0)$ and $B = J_2(0) \oplus J_2(0)$ are not similar to each other, but have $p_A(x) = p_B(x) = x^4$, $q_A(x) = q_B(x) = x^2$.
4. The Jordan canonical form of A is not necessarily determined by the eigenvalues and the algebraic multiplicity, geometric multiplicity, and index of each eigenvalue. For example, the Jordan matrices $A = J_3(0) \oplus J_3(0) \oplus J_1(0)$ and $B = J_3(0) \oplus J_2(0) \oplus J_2(0)$ are not similar to each other, but have $\alpha_A(0) = \alpha_B(0) = 7$, $\gamma_A(0) = \gamma_B(0) = 3$, $\nu_0(A) = \nu_0(B) = 3$ (and $p_A(x) = p_B(x) = x^7$, $q_A(x) = q_B(x) = x^3$).

TABLE 6.1 $\text{rank}(A - \lambda I)^k$

$k =$	1	2	3	4	5
$\lambda = 1$	11	10	9	9	9
$\lambda = 2$	12	10	10	10	10
$\lambda = 3$	12	11	10	9	9

5. We use Algorithm 1 to find a matrix C such that $C^{-1}AC = J_A$ for

$$A = \begin{bmatrix} -2 & 3 & 0 & 1 & -1 \\ 4 & 0 & 3 & 0 & -2 \\ 6 & -3 & 3 & -1 & -1 \\ -8 & 6 & -3 & 2 & 0 \\ 2 & 3 & 3 & 1 & -3 \end{bmatrix}. \text{ Computations show that } p_A(x) = x^5 \text{ and } \ker A = \text{Span}(\mathbf{z}_1, \mathbf{z}_2, \mathbf{z}_3),$$

where $\mathbf{z}_1 = [3, 2, -4, 0, 0]^T$, $\mathbf{z}_2 = [0, 1, 0, -3, 0]^T$, $\mathbf{z}_3 = [3, 4, 0, 0, 6]^T$.

For Step 1, $A^3 = \mathbf{0}$, and $\text{range}(A^2) = \text{Span}(\mathbf{b}_1)$ where $\mathbf{b}_1 = [-1, -1, 0, -1, -2]^T$.

Then $\mathcal{B} = \{\mathbf{b}_1, \mathbf{z}_1, \mathbf{z}_2\}$ is a suitable basis (any 2 of $\{\mathbf{z}_1, \mathbf{z}_2, \mathbf{z}_3\}$ will work in this case).

For Step 2, construct a Jordan chain above $\mathbf{b}_1$ by solving $A^2\mathbf{u}_1 = \mathbf{b}_1$. There are many possible solutions; we choose $\mathbf{u}_1 = [0, 0, 0, 0, 1]^T$. Then $A\mathbf{u}_1 = [-1, -2, -1, 0, -3]^T$,

$$C = \begin{bmatrix} -1 & -1 & 0 & 3 & 0 \\ -1 & -2 & 0 & 2 & 1 \\ 0 & -1 & 0 & -4 & 0 \\ -1 & 0 & 0 & 0 & -3 \\ -2 & -3 & 1 & 0 & 0 \end{bmatrix}, \quad \text{and} \quad J_A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \oplus [0] \oplus [0].$$

6. We compute the Jordan canonical form of a 14×14 matrix A by the method in Fact 3, where the necessary data about the eigenvalues of A and ranks is given in Table 6.1.

- $\lambda = 1$ – The number of blocks of size 1 is $14 + 10 - 2 \cdot 11 = 2$.
 – The number of blocks of size 2 is $11 + 9 - 2 \cdot 10 = 0$.
 – The number of blocks of size 3 is $10 + 9 - 2 \cdot 9 = 1$.

So $v_1 = 3$ and $b_1 = 3$.

- $\lambda = 2$ – The number of blocks of size 1 is $14 + 10 - 2 \cdot 12 = 0$.
 – The number of blocks of size 2 is $12 + 10 - 2 \cdot 10 = 2$.

So, $v_2 = 2$ and $b_2 = 2$.

- $\lambda = 3$ – The number of blocks of size 1 is $14 + 11 - 2 \cdot 12 = 1$.
 – The number of blocks of size 2 is $12 + 10 - 2 \cdot 11 = 0$.
 – The number of blocks of size 3 is $11 + 9 - 2 \cdot 10 = 0$.
 – The number of blocks of size 4 is $10 + 9 - 2 \cdot 9 = 1$.

So, $v_3 = 4$ and $b_3 = 2$.

From this information,

$$J_A = J_3(1) \oplus J_1(1) \oplus J_1(1) \oplus J_2(2) \oplus J_2(2) \oplus J_4(3) \oplus J_1(3).$$

6.3 Real-Jordan Canonical Form

The real-Jordan canonical form is used in applications to differential equations, dynamical systems, and control theory (see Chapter 56). The real-Jordan canonical form is discussed here only for matrices and with limited discussion of generalized eigenspaces; more generality is possible, and is readily derivable from the corresponding results for the Jordan canonical form.

Definitions:

Let $A \in \mathbb{R}^{n \times n}$, $\alpha, \beta, \alpha_j, \beta_j \in \mathbb{R}$.

The **real generalized eigenspace** of A at eigenvalue $\alpha + \beta i$ is

$$E(A, \alpha + \beta i) = \begin{cases} \ker((A^2 - 2\alpha A + (\alpha^2 + \beta^2)I)^\nu) & \text{if } \beta \neq 0 \\ N_\alpha^\nu(A) = \ker((A - \alpha I)^\nu) & \text{if } \beta = 0. \end{cases}$$

The vector $\mathbf{x} \in \mathbb{R}^n$ is a **real generalized eigenvector** of A for $\alpha + \beta i$ if $\mathbf{x} \neq \mathbf{0}$ and $\mathbf{x} \in E(A, \alpha + \beta i)$.

For $\alpha, \beta \in \mathbb{R}$ with $\beta \neq 0$, and even positive integer $2k$, the **real-Jordan block** of size $2k$ with eigenvalue $\alpha + \beta i$ is the $2k \times 2k$ matrix having k copies of $M_2(\alpha, \beta) = \begin{bmatrix} \alpha & \beta \\ -\beta & \alpha \end{bmatrix}$ on the (block matrix) diagonal, $k-1$ copies of $I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ on the first (block matrix) superdiagonal, and copies of $0_2 = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ everywhere else, i.e.,

$$J_{2k}^{\mathbb{R}}(\alpha + \beta i) = \begin{bmatrix} M_2(\alpha, \beta) & I_2 & 0_2 & \cdots & 0_2 \\ 0_2 & M_2(\alpha, \beta) & I_2 & \cdots & 0_2 \\ \vdots & \ddots & \ddots & & \vdots \\ 0_2 & \cdots & 0_2 & M_2(\alpha, \beta) & I_2 \\ 0_2 & \cdots & 0_2 & 0_2 & M_2(\alpha, \beta) \end{bmatrix}.$$

A **real-Jordan matrix** (or a **matrix in real-Jordan canonical form**) is a block diagonal matrix having diagonal blocks that are Jordan blocks or real-Jordan blocks, i.e., a matrix of the form $J_{m_1}(\alpha_1) \oplus \cdots \oplus J_{m_t}(\alpha_t) \oplus J_{2k_{t+1}}^{\mathbb{R}}(\alpha_{t+1} + \beta_{t+1}i) \oplus \cdots \oplus J_{2k_s}^{\mathbb{R}}(\alpha_s + \beta_s i)$ (or a permutation of the direct summands).

A **real-Jordan canonical form** of matrix A , denoted $J_A^{\mathbb{R}}$ or $\text{JCF}^{\mathbb{R}}(A)$, is a real-Jordan matrix that is similar to A . It is conventional to use $\beta_j > 0$, to group the blocks for the same eigenvalue together, and to order the Jordan blocks with the same eigenvalue in nonincreasing size order.

The **total number of Jordan blocks** in a real-Jordan canonical form of A is the number of blocks (Jordan or real-Jordan) in $J_A^{\mathbb{R}}$.

Facts:

Facts requiring proof for which no specific reference is given can be found in [HJ85, Chapter 3].

Notation: $A, B \in \mathbb{R}^{n \times n}$, $\alpha, \beta, \alpha_j, \beta_j \in \mathbb{R}$.

1. The real generalized eigenspace of a complex number $\lambda = \alpha + \beta i$ and its conjugate $\bar{\lambda} = \alpha - \beta i$ are equal, i.e., $E(A, \alpha + \beta i) = E(A, \alpha - \beta i)$.
2. The real-Jordan blocks with a nonreal complex number and its conjugate are similar to each other.
3. A has a real-Jordan canonical form $J_A^{\mathbb{R}}$, and $J_A^{\mathbb{R}}$ is unique up to permutation of the diagonal (real-) Jordan blocks.
4. A, B are similar if and only if their real-Jordan canonical forms have the same set of Jordan and real-Jordan blocks (although the order may vary).
5. If all the eigenvalues of A are real, then $J_A^{\mathbb{R}}$ is the same as J_A (up to the order of the Jordan blocks).
6. The real-Jordan canonical form of A can be computed from the Jordan canonical form of A . The nonreal eigenvalues occur in conjugate pairs, and if $\beta > 0$, then each size k Jordan block with eigenvalue $\alpha + \beta i$ can be paired with a size k Jordan block for $\alpha - \beta i$. Then $J_k(\alpha + \beta i) \oplus J_k(\alpha - \beta i)$ is replaced by $J_{2k}^{\mathbb{R}}(\alpha + \beta i)$. The Jordan blocks of $J_A^{\mathbb{R}}$ with real eigenvalues are the same as the those of J_A .
7. The total number of Jordan and real-Jordan blocks in a real-Jordan canonical form of A is the number of Jordan blocks with a real eigenvalue plus half the number of Jordan blocks with a nonreal eigenvalue in a Jordan canonical form of A .

8. If $\beta \neq 0$, the size of the largest real-Jordan block with eigenvalue $\alpha + \beta i$ is twice the multiplicity of $x^2 - 2\alpha x + (\alpha^2 + \beta^2)$ as a factor of the minimal polynomial $q_A(x)$ of A .
9. If $\beta \neq 0$, the sum of the sizes of all the real-Jordan blocks with eigenvalue $\alpha + \beta i$ in J_A equals twice the algebraic multiplicity $\alpha_A(\alpha + \beta i)$.
10. If $\beta \neq 0$, $\dim E(A, \alpha + \beta i) = \alpha_A(\alpha + \beta i)$.
11. If $A = A_1 \oplus \cdots \oplus A_m$, then $J_{A_1}^{\mathbb{R}} \oplus \cdots \oplus J_{A_m}^{\mathbb{R}}$ is a real-Jordan canonical form of A .

Examples:

1. Let $a = \begin{bmatrix} -10 & 6 & -4 & 4 & 0 \\ -17 & 10 & -4 & 6 & -1 \\ -4 & 2 & -3 & 2 & 1 \\ -11 & 6 & -11 & 6 & 3 \\ -4 & 2 & -4 & 2 & 2 \end{bmatrix}$. Since the characteristic and minimal polynomials of A are

$$\text{both } x^5 - 5x^4 + 12x^3 - 16x^2 + 12x - 4 = (x - 1)(x^2 - 2x + 2)^2,$$

$$J_A^{\mathbb{R}} = \begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ -1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

2. The requirement that $\beta \neq 0$ is important. $A = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ is *not* a real-Jordan matrix;

$$J_A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

6.4 Rational Canonical Form: Elementary Divisors

The elementary divisors rational canonical form is closely related to the Jordan canonical form (see Fact 7 below). A rational canonical form (either the elementary divisors or the invariant factors version, cf. Section 6.6) is used when it is desirable to stay within a field that is not algebraically closed, such as the rational numbers.

Definitions:

Let F be a field.

For a monic polynomial $p(x) = x^n + c_{n-1}x^{n-1} + \cdots + c_2x^2 + c_1x + c_0 \in F[x]$ (with $n \geq 1$), the

companion matrix of $p(x)$ is the $n \times n$ matrix $C(p) = \begin{bmatrix} 0 & 0 & \cdots & -c_0 \\ 1 & 0 & \cdots & -c_1 \\ \vdots & \ddots & & \vdots \\ 0 & \cdots & 1 & -c_{n-1} \end{bmatrix}$.

An **elementary divisors rational canonical form matrix (ED-RCF matrix)** (over F) is a block diagonal matrix of the form $C(h_1^{m_1}) \oplus \cdots \oplus C(h_t^{m_t})$ where each $h_i(x)$ is a monic polynomial that is irreducible over F .

The **elementary divisors** of the ED-RCF matrix $C(h_1^{m_1}) \oplus \cdots \oplus C(h_t^{m_t})$ are the polynomials $h_i(x)^{m_i}$, $i = 1, \dots, t$.

An **elementary divisors rational canonical form** of matrix $A \in F^{n \times n}$, denoted $\text{RCF}_{ED}(A)$, is an ED-RCF matrix that is similar to A . It is conventional to group the companion matrices associated with powers of the same irreducible polynomial together, and within such a group to order the blocks in size order.

The **elementary divisors** of A are the elementary divisors of $\text{RCF}_{ED}(A)$.

Let V be a finite dimensional vector space over F , and let T be a linear operator on V .

An **ED-RCF basis** for T is an ordered basis $\mathcal{B}$ of V , with respect to which the matrix ${}_B[T]_B$ of T is an ED-RCF matrix. In this case, ${}_B[T]_B$ is an elementary divisors rational canonical form of T , denoted $\text{RCF}_{ED}(T)$, and the **elementary divisors** of T are the elementary divisors of $\text{RCF}_{ED}(T) = {}_B[T]_B$.

Let $q(x)$ be a monic polynomial over F .

A **primary decomposition** of a nonconstant monic polynomial $q(x)$ over F is a factorization $q(x) = (h_1(x))^{m_1} \cdots (h_r(x))^{m_r}$, where the $h_i(x)$, $i = 1, \dots, r$ are distinct monic irreducible polynomials over F .

The factors $(h_i(x))^{m_i}$ in a primary decomposition of $q(x)$ are the **primary factors** of $q(x)$.

Facts:

Facts requiring proof for which no specific reference is given can be found in [HK71, Chapter 7] or [DF04, Chapter 12].

1. The characteristic and minimal polynomials of the companion matrix $C(p)$ are both equal to $p(x)$.
2. Whether or not a matrix is an ED-RCF matrix depends on polynomial irreducibility, which depends on the field. See Example 1 below.
3. Every matrix $A \in F^{n \times n}$ is similar to an ED-RCF matrix, $\text{RCF}_{ED}(A)$, and $\text{RCF}_{ED}(A)$ is unique up to permutation of the companion matrix blocks on the diagonal. In particular, the elementary divisors of A are uniquely determined by A .
4. $A, B \in F^{n \times n}$ are similar if and only if they have the same elementary divisors.
5. (See Fact 3 in Section 6.2) For $A \in F^{n \times n}$, the elementary divisors and, hence, $\text{RCF}_{ED}(A)$ can be found from the irreducible factors $h_i(x)$ of the characteristic polynomial of A and the ranks of powers of $h_i(A)$. Specifically, the number of times $h_i(x)^k$ appears as an elementary divisor of A is

$$\frac{1}{\deg h_i(x)} (\text{rank}(h_i(A))^{k-1} + \text{rank}(h_i(A))^{k+1} - 2 \text{rank}(h_i(A))^k).$$

6. If $A \in F^{n \times n}$ has n eigenvalues in F , then the elementary divisors of A are the polynomials $(x - \lambda)^k$, where the $J_k(\lambda)$ are the Jordan blocks of J_A .
7. There is a natural association between the diagonal blocks in the elementary divisors rational canonical form of A and the Jordan canonical form of A in $\hat{F}^{n \times n}$, where $\hat{F}$ is the algebraic closure of F . Let $h(x)^m$ be an elementary divisor of A , and factor $h(x)$ into monic linear factors over $\hat{F}$, $h(x) = (x - \lambda_1) \cdots (x - \lambda_t)$. If the roots of $h(x)$ are distinct (e.g., if the characteristic of F is 0 or F is a finite field), then the ED-RCF diagonal block $C(h^m)$ is associated with the Jordan blocks $J_m(\lambda_i)$, $i = 1, \dots, t$. If the characteristic of F is p and $h(x)$ has repeated roots, then all roots have the same multiplicity p^k (for some positive integer k) and the ED-RCF diagonal block $C(h^m)$ is associated with the Jordan blocks $J_{p^k m}(\lambda_i)$, $i = 1, \dots, t$.
8. [HK71, Chapter 4.5] Every monic polynomial $q(x)$ over F has a primary decomposition. The primary decomposition is unique up to the order of the monic irreducible polynomials, i.e., the set of primary factors of $q(x)$ is unique.
9. [HK71, Chapter 6.8] Let $q(x) \in F[x]$, let $h_i(x)^{m_i}$, $i = 1, \dots, r$ be the primary factors, and define $f_i(x) = \frac{q(x)}{h_i(x)^{m_i}}$. Then there exist polynomials $g_i(x)$ such that $f_1(x)g_1(x) + \cdots + f_r(x)g_r(x) = 1$. Let $A \in F^{n \times n}$ and let $q_A(x) = (h_1(x))^{m_1} \cdots (h_r(x))^{m_r}$ be a primary decomposition of its minimal polynomial.
10. Every primary factor $h_i(x)^{m_i}$ of $q_A(x)$ is an elementary divisor of A .
11. Every elementary divisor of A is of the form $(h_i(x))^m$ with $m \leq m_i$ for some $i \in \{1, \dots, r\}$.

12. [HK71, Chapter 6.8] *Primary Decomposition Theorem*

- (a) $F^n = \ker(h_1(A)^{m_1}) \oplus \cdots \oplus \ker(h_r(A)^{m_r})$.
- (b) Let f_i and g_i be as defined in Fact 9. Then for $i = 1, \dots, r$, $E_i = f_i(A)g_i(A)$ is the projection onto $\ker(h_i(A)^{m_i})$ along $\ker(h_1(A)^{m_1}) \oplus \cdots \oplus \ker(h_{i-1}(A)^{m_{i-1}}) \oplus \ker(h_{i+1}(A)^{m_{i+1}}) \oplus \cdots \oplus \ker(h_r(A)^{m_r})$.
- (c) The $E_i = f_i(A)g_i(A)$ are mutually orthogonal idempotents (i.e., $E_i^2 = E_i$ and $E_i E_j = 0$ if $i \neq j$) and $I = E_1 + \cdots + E_r$.

13. [HK71, Chapter 6.7] If $A \in F^{n \times n}$ is diagonalizable, then $A = \mu_1 E_1 + \cdots + \mu_r E_r$ where the E_i are the projections defined in Fact 12 with primary factors $h_i(x)^{m_i} = (x - \mu_i)$ of $q_A(x)$.

Let V be an n -dimensional vector space over F , and let T be a linear operator on V .

14. Facts 3, 5 to 7, and 10 to 13 remain true when matrix A is replaced by linear operator T ; in particular, $\text{RCF}_{ED}(T)$ exists and is independent (up to permutation of the companion matrix diagonal blocks) of the ordered basis of V used to compute it, and the elementary divisors of T are independent of basis.
15. If $\hat{T}$ denotes T restricted to $\ker(h_i(T)^{m_i})$, then the minimal polynomial of $\hat{T}$ is $h_i(T)^{m_i}$.

Examples:

1. Let $A = [-1] \oplus [-1] \oplus \begin{bmatrix} 0 & 0 & -1 \\ 1 & 0 & -3 \\ 0 & 1 & -3 \end{bmatrix} \oplus \begin{bmatrix} 0 & 2 \\ 1 & 0 \end{bmatrix} \oplus \begin{bmatrix} 0 & 0 & 0 & -4 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 4 \\ 0 & 0 & 1 & 0 \end{bmatrix}$. Over $\mathbb{Q}$, A is an ED-RCF

matrix and its elementary divisors are $x + 1, x + 1, (x + 1)^3, x^2 - 2, (x^2 - 2)^2$. A is not an ED-RCF matrix over $\mathbb{C}$ because $x^2 - 2$ is not irreducible over $\mathbb{C}$.

$$\text{JCF}(A) = [-1] \oplus [-1] \oplus \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & -1 \end{bmatrix} \oplus [\sqrt{2}] \oplus [-\sqrt{2}] \oplus \begin{bmatrix} \sqrt{2} & 1 \\ 0 & \sqrt{2} \end{bmatrix} \oplus \begin{bmatrix} -\sqrt{2} & 1 \\ 0 & -\sqrt{2} \end{bmatrix},$$

where the order of the Jordan blocks has been chosen to emphasize the connection to $\text{RCF}_{ED}(A) = A$.

2. Let $A = \begin{bmatrix} -2 & 2 & -2 & 1 & 1 \\ 6 & -2 & 2 & -2 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ -12 & 7 & -8 & 5 & 4 \\ 0 & 0 & -1 & 0 & 2 \end{bmatrix} \in \mathbb{Q}^{5 \times 5}$. We use Fact 5 to determine the elementary divisors

rational canonical form of A . The following computations can be performed easily over $\mathbb{Q}$ in a computer algebra system such as Mathematica, Maple, or MATLAB[®] (see Chapters 71, 72, 73), or on a matrix-capable calculator. $p_A(x) = x^5 - 3x^4 + x^3 + 5x^2 - 6x + 2 = (x - 1)^3(x^2 - 2)$. Table 6.2 gives the of ranks $h_i(A)^k$ where $h_i(x)$ is shown in the left column.

$h(x) = x - 1$ The number of times $x - 1$ appears as an elementary divisor is $5 + 2 - 2 \cdot 3 = 1$.

The number of $(x - 1)^2$ appears as an elementary divisor is $3 + 2 - 2 \cdot 2 = 1$.

$h(x) = x^2 - 2$ The number of times $x^2 - 2$ appears as an elementary divisor is $(5 + 3 - 2 \cdot 3)/2 = 1$.

$$\text{Thus, } \text{RCF}_{ED}(A) = C(x - 1) \oplus C((x - 1)^2) \oplus C(x^2 - 2) = [1] \oplus \begin{bmatrix} 0 & -1 \\ 1 & 2 \end{bmatrix} \oplus \begin{bmatrix} 0 & 2 \\ 1 & 0 \end{bmatrix}.$$

TABLE 6.2 $\text{rank}(h(A)^k)$

$k =$	1	2	3
$h_1(x) = x - 1$	3	2	2
$h_2(x) = x^2 - 2$	3	3	3

3. We find the projections $E_i, i = 1, 2$ in Fact 12 for A in the previous example. From the elementary divisors of A , $q_A(x) = (x - 1)^2(x^2 - 2)$. Let $h_1(x) = (x - 1)^2, h_2(x) = x^2 - 2$. Then $f_1(x) = x^2 - 2, f_2(x) = (x - 1)^2$. *Note:* normally the $f_i(x)$ will not be primary factors; this happens here because there are only two primary factors. If we choose $g_1(x) = -(2x - 1), g_2(x) = 2x + 3$, then $1 = f_1(x)g_1(x) + f_2(x)g_2(x)$ (g_1, g_2 can be found by the Euclidean algorithm). Then

$$E_1 = f_1(A)g_1(A) = \begin{bmatrix} -2 & 1 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ -6 & 3 & -2 & 3 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \quad \text{and} \quad E_2 = f_2(A)g_2(A) = \begin{bmatrix} 3 & -1 & 1 & -1 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 6 & -3 & 2 & -2 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix},$$

and it is easy to verify that $E_1^2 = E_1, E_2^2 = E_2, E_1E_2 = E_2E_1 = 0$, and $E_1 + E_2 = I$.

6.5 Smith Normal Form on $F[x]^{n \times n}$

For a matrix $A \in F^{n \times n}$, the Smith normal form of $xI_n - A$ is an important tool for the computation of the invariant factors rational canonical form of A discussed in Section 6.6. In this section, Smith normal form is discussed only for matrices in $F[x]^{n \times n}$, and the emphasis is on finding the Smith normal form of $xI_n - A$, where $A \in F^{n \times n}$. Smith normal form is used more generally for matrices over principal ideal domains (see Section 23.2); it is not used extensively as a canonical form within $F^{n \times n}$, since the Smith normal form of a matrix $A \in F^{n \times n}$ of rank k is $I_k \oplus 0_{n-k}$.

Definitions:

Let F be a field. For $M \in F[x]^{n \times n}$, the following operations are the **elementary row and column operations** on M :

- Interchange rows i, j , denoted $R_i \leftrightarrow R_j$ (analogous column operation denoted $C_i \leftrightarrow C_j$).
- Add a $p(x)$ multiple of row j to row i , denoted $R_i + p(x)R_j \rightarrow R_i$ (analogous column operation denoted $C_i + p(x)C_j \rightarrow C_i$).
- Multiply row i by a nonzero element b of F , denoted $bR_i \rightarrow R_i$ (analogous column operation denoted $bC_i \rightarrow C_i$).

A **Smith normal matrix** in $F[x]^{n \times n}$ is a diagonal matrix $D = \text{diag}(1, \dots, 1, a_1(x), \dots, a_s(x), 0, \dots, 0)$, where the $a_i(x)$ are monic nonconstant polynomials such that $a_i(x)$ divides $a_{i+1}(x)$ for $i = 1, \dots, s - 1$.

The **Smith normal form** of $M \in F[x]^{n \times n}$ is the Smith normal matrix obtained from M by elementary row and column operations.

For $A \in F^{n \times n}$, the monic nonconstant polynomials of the Smith normal form of $xI_n - A$ are the **Smith invariant factors** of A .

Facts:

Facts requiring proof for which no specific reference is given can be found in [HK71, Chapter 7] or [DF04, Chapter 12].

- Let $M \in F[x]^{n \times n}$. Then M has a unique Smith normal form.
- Let $A \in F^{n \times n}$. There are no zeros on the diagonal of the Smith normal form of $xI_n - A$.
- (*Division Property*) If $a(x), b(x) \in F[x]$ and $b(x) \neq 0$, then there exist polynomials $q(x), r(x)$ such that $a(x) = q(x)b(x) + r(x)$ and $r(x) = 0$ or $\deg r(x) < \deg b(x)$.
- The Smith normal form of $M = xI - A$ and, thus, the Smith invariant factors of A can be computed as follows:

- For $k = 1, \dots, n - 1$
 - Use elementary row and column operations and the division property of $F[x]$ to place the greatest common divisor of the entries of $M[\{k, \dots, n\}]$ in the k th diagonal position.
 - Use elementary row and column operations to create zeros in all nondiagonal positions in row k and column k .
- Make the n th diagonal entry monic by multiplying the last column by a nonzero element of F .

This process is illustrated in Example 1 below.

Examples:

1. Let $A = \begin{bmatrix} 1 & 1 & 1 & -1 \\ 0 & 3 & 2 & -2 \\ 2 & 0 & 4 & -2 \\ 4 & 0 & 6 & -3 \end{bmatrix}$. We use the method in Fact 4 above to find the Smith normal form of

$M = xI - A$ and invariant factors of A .

- $k = 1$: Use the row and column operations on M (in the order shown):

$$R_1 \leftrightarrow R_3, -\frac{1}{2}R_1 \rightarrow R_1, R_3 + (1-x)R_1 \rightarrow R_3, R_4 + 4R_1 \rightarrow R_4,$$

$$C_3 + (-2 + \frac{x}{2})C_1 \rightarrow C_3, C_4 + C_1 \rightarrow C_4$$

$$\text{to obtain } M_1 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & x-3 & -2 & 2 \\ 0 & -1 & \frac{x^2}{2} - \frac{5x}{2} + 1 & x \\ 0 & 0 & 2-2x & x-1 \end{bmatrix}.$$

- $k = 2$: Use the row and column operations on M_1 (in the order shown):

$$R_3 \leftrightarrow R_2, -1R_2 \rightarrow R_2, R_3 + (3-x)R_2 \rightarrow R_3,$$

$$C_3 + (1 - \frac{5x}{2} + \frac{x^2}{2})C_2 \rightarrow C_3, C_4 + xC_2 \rightarrow C_4$$

$$\text{to obtain } M_2 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \frac{x^3}{2} - 4x^2 + \frac{17x}{2} - 5 & x^2 - 3x + 2 \\ 0 & 0 & 2-2x & x-1 \end{bmatrix}.$$

- $k = 3$ (and final step): Use the row and column operations on M_2 (in the order shown):

$$R_3 \leftrightarrow R_4, -\frac{1}{2}R_3 \rightarrow R_3, R_4 + \frac{-1}{2}(x-2)(x-5)R_3 \rightarrow R_4,$$

$$C_4 + \frac{1}{2}C_3 \rightarrow C_4, 4C_4 \rightarrow C_4$$

$$\text{to obtain the Smith normal form of } M, M_3 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & x-1 & 0 \\ 0 & 0 & 0 & x^3 - 4x^2 + 5x - 2 \end{bmatrix}.$$

The Smith invariant factors of A are $x - 1, x^3 - 4x^2 + 5x - 2$.

6.6 Rational Canonical Form: Invariant Factors

Like the elementary divisors version, the invariant factors rational canonical form does not require the field to be algebraically closed. It has two other advantages: This canonical form is unique (not just unique up to permutation), and (unlike elementary divisors rational canonical form) whether a matrix is in invariant factors rational canonical form is independent of the field (see Fact 2 below).

Definitions:

Let F be a field. An **invariant factors rational canonical form matrix (IF-RCF matrix)** is a block diagonal matrix of the form $C(a_1) \oplus \cdots \oplus C(a_s)$, where $a_i(x)$ divides $a_{i+1}(x)$ for $i = 1, \dots, s-1$.

The **invariant factors** of the IF-RCF matrix $C(a_1) \oplus \cdots \oplus C(a_s)$ are the polynomials $a_i(x)$, $i = 1, \dots, s$.

The **invariant factors rational canonical form** of matrix $A \in F^{n \times n}$, denoted $\text{RCF}_{IF}(A)$, is the IF-RCF matrix that is similar to A .

The **invariant factors** of A are the invariant factors of $\text{RCF}_{IF}(A)$.

Let V be a finite dimensional vector space over F and let T be a linear operator on V .

An **IF-RCF basis** for T is an ordered basis $\mathcal{B}$ of V , with respect to which the matrix ${}_B[T]_B$ of T is an IF-RCF matrix. In this case, ${}_B[T]_B$ is the **invariant factors rational canonical form** of T , denoted $\text{RCF}_{IF}(T)$, and the **invariant factors** of T are the invariant factors of $\text{RCF}_{IF}(T) = {}_B[T]_B$.

Facts:

Facts requiring proof for which no specific reference is given can be found in [HK71, Chapter 7] or [DF04, Chapter 12]. *Notation:* $A \in F^{n \times n}$.

1. Every square matrix A is similar to a unique IF-RCF matrix, $\text{RCF}_{IF}(A)$.
2. $\text{RCF}_{IF}(A)$ is independent of field. That is, if K is an extension field of F and A is considered as an element of $K^{n \times n}$, $\text{RCF}_{IF}(A)$ is the same as when A is considered as an element of $F^{n \times n}$.
3. Let $B \in F^{n \times n}$. Then A, B are similar if and only if $\text{RCF}_{IF}(A) = \text{RCF}_{IF}(B)$.
4. The characteristic polynomial is the product of the invariant factors of A , i.e., $p_A(x) = a_1(x) \cdots a_s(x)$.
5. The minimal polynomial of A is the invariant factor of highest degree, i.e., $q_A(x) = a_s(x)$.
6. The elementary divisors of $A \in F^{n \times n}$ are the primary factors (over F) of the invariant factors of A .
7. The Smith invariant factors of A are the invariant factors of A .
8. [DF04, Chapter 12.2] $\text{RCF}_{IF}(A)$ and a nonsingular matrix $S \in F^{n \times n}$ such that $S^{-1}AS = \text{RCF}_{IF}(A)$ can be computed by Algorithm 2.

Algorithm 2: Rational Canonical Form (invariant factors)

1. Compute the Smith normal form D of $M = xI - A$ as in Fact 4 of section 6.5, keeping track of the elementary row operations, in the order performed (column operations need not be recorded).
2. The invariant factors are the nonconstant diagonal elements $a_1(x), \dots, a_s(x)$ of D .
3. Let $d_1, \dots, d_s$ denote the degrees of $a_1(x), \dots, a_s(x)$.
4. Let $G = I$.
5. FOR $k = 1, \dots$, number of row operations performed in step 1
 - (a) If the k th row operation is $R_i \leftrightarrow R_j$, then perform column operation $C_j \leftrightarrow C_i$ on G .
 - (b) If the k th row operation is $R_i + p(x)R_j \rightarrow R_i$, then perform column operation $C_j - p(A)C_i \rightarrow C_j$ on G (*note index reversal*).
 - (c) If the k th row operation is $bR_i \rightarrow R_i$, then perform column operation $\frac{1}{b}C_i \rightarrow C_i$ on G .
6. G will have 0s in the first $n - s$ columns; denote the remaining columns of G by $\mathbf{g}_1, \dots, \mathbf{g}_s$.
7. Initially S has no columns.
8. FOR $k = 1, \dots, s$
 - (a) Insert $\mathbf{g}_k$ as the next column of S (working left to right).
 - (b) FOR $i = 1, \dots, d_k - 1$
 - Insert A times the last column inserted as the next column of S .
9. $\text{RCF}_{IF}(A) = S^{-1}AS$.

9. Let V be an n -dimensional vector space over F , and let T be a linear operator on V . Facts 1, 2, 4 to 6 remain true when matrix A is replaced by linear operator T ; in particular, $\text{RCF}_{IF}(T)$ exists and is unique (independent of the ordered basis of V used to compute it).

Examples:

1. We can use the elementary divisors already computed to find the invariant factors and IF-RCF of A in Example 2 of Section 6.4. The elementary divisors of A are $x - 1$, $(x - 1)^2$, $x^2 - 2$. We combine these, working down from the highest power of each irreducible polynomial.
 $a_2(x) = (x - 1)^2(x^2 - 2) = x^4 - 2x^3 - x^2 + 4x - 2$, $a_1(x) = x - 1$. Then

$$\text{RCF}_{IF}(A) = C(x - 1) \oplus C(x^4 - 2x^3 - x^2 + 4x - 2) = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 \\ 0 & 1 & 0 & 0 & -4 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 2 \end{bmatrix}.$$

2. By Fact 7, for the matrix A in Example 1 in Section 6.5, $\text{RCF}_{IF}(A) = C(x - 1) \oplus C(x^3 - 4x^2 + 5x - 2)$.
 3. We can use Algorithm 2 to find a matrix S such that $\text{RCF}_{IF}(A) = S^{-1}AS$ for the matrix A in Example 1.

- $k = 1$: Starting with $G = I_4$, perform the column operations (in the order shown):

$$C_1 \leftrightarrow C_3, -2C_1 \rightarrow C_1, C_1 - (I_4 - A)C_3 \rightarrow C_1, C_1 - 4C_4 \rightarrow C_1,$$

$$\text{to obtain } G_1 = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

- $k = 2$: Use column operations on G (in the order shown):

$$C_3 \leftrightarrow C_2, -1C_2 \rightarrow C_2, C_2 - (3I_4 - A)C_3 \rightarrow C_2,$$

$$\text{to obtain } G = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

- $k = 3$ (and final step of Fact 4 in Section 6.5):

Use column operations on G (in the order shown):

$$C_3 \leftrightarrow C_4, -2C_3 \rightarrow C_3, C_3 + \frac{1}{2}(A - 2I_4)(A - 5I_4)C_4 \rightarrow C_3,$$

$$\text{to obtain } G = [\mathbf{g}_1, \mathbf{g}_2, \mathbf{g}_3, \mathbf{g}_4] = \begin{bmatrix} 0 & 0 & -\frac{3}{2} & 0 \\ 0 & 0 & -1 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

$$\text{Then } S = [\mathbf{g}_3, \mathbf{g}_4, A\mathbf{g}_4, A^2\mathbf{g}_4] = \begin{bmatrix} -\frac{3}{2} & 0 & 1 & 4 \\ -1 & 1 & 3 & 9 \\ 1 & 0 & 0 & 2 \\ 0 & 0 & 0 & 4 \end{bmatrix} \quad \text{and} \quad \text{RCF}_{IF}(A) = S^{-1}AS = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 \\ 0 & 1 & 0 & -5 \\ 0 & 0 & 1 & 4 \end{bmatrix}.$$

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7

Unitary Similarity, Normal Matrices, and Spectral Theory

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Unitary transformations preserve the inner product. Hence, they preserve the metric quantities that stem from the inner product, such as length, distance, and angle. While a general similarity preserves the algebraic features of a linear transformation, such as the characteristic and minimal polynomials, the rank, and the Jordan canonical form, unitary similarities also preserve metric features such as the norm, singular values, and the numerical range. Unitary similarities are desirable in computational linear algebra for stability reasons.

Normal transformations are those which have an orthogonal basis of eigenvectors and, thus, can be represented by diagonal matrices relative to an orthonormal basis. The class of normal transformations includes Hermitian, skew-Hermitian, and unitary transformations; studying normal matrices leads to a more unified understanding of all of these special types of transformations. Often, results that are discovered first for Hermitian matrices can be generalized to the class of normal matrices. Since normal matrices are unitarily similar to diagonal matrices, things that are obviously true for diagonal matrices often hold for normal matrices as well; for example, the singular values of a normal matrix are the absolute values of the eigenvalues. Normal matrices have two important properties — diagonalizability and an orthonormal basis of eigenvectors — that tend to make life easier in both theoretical and computational situations.

7.1 Unitary Similarity

In this subsection, all matrices are over the complex numbers and are square. All vector spaces are finite dimensional complex inner product spaces.

Definitions:

A matrix U is **unitary** if $U^*U = I$.

A matrix Q is **orthogonal** if $Q^T Q = I$.

Note: This extends the definition of orthogonal matrix given earlier in Section 5.2 for real matrices.